technica engineering

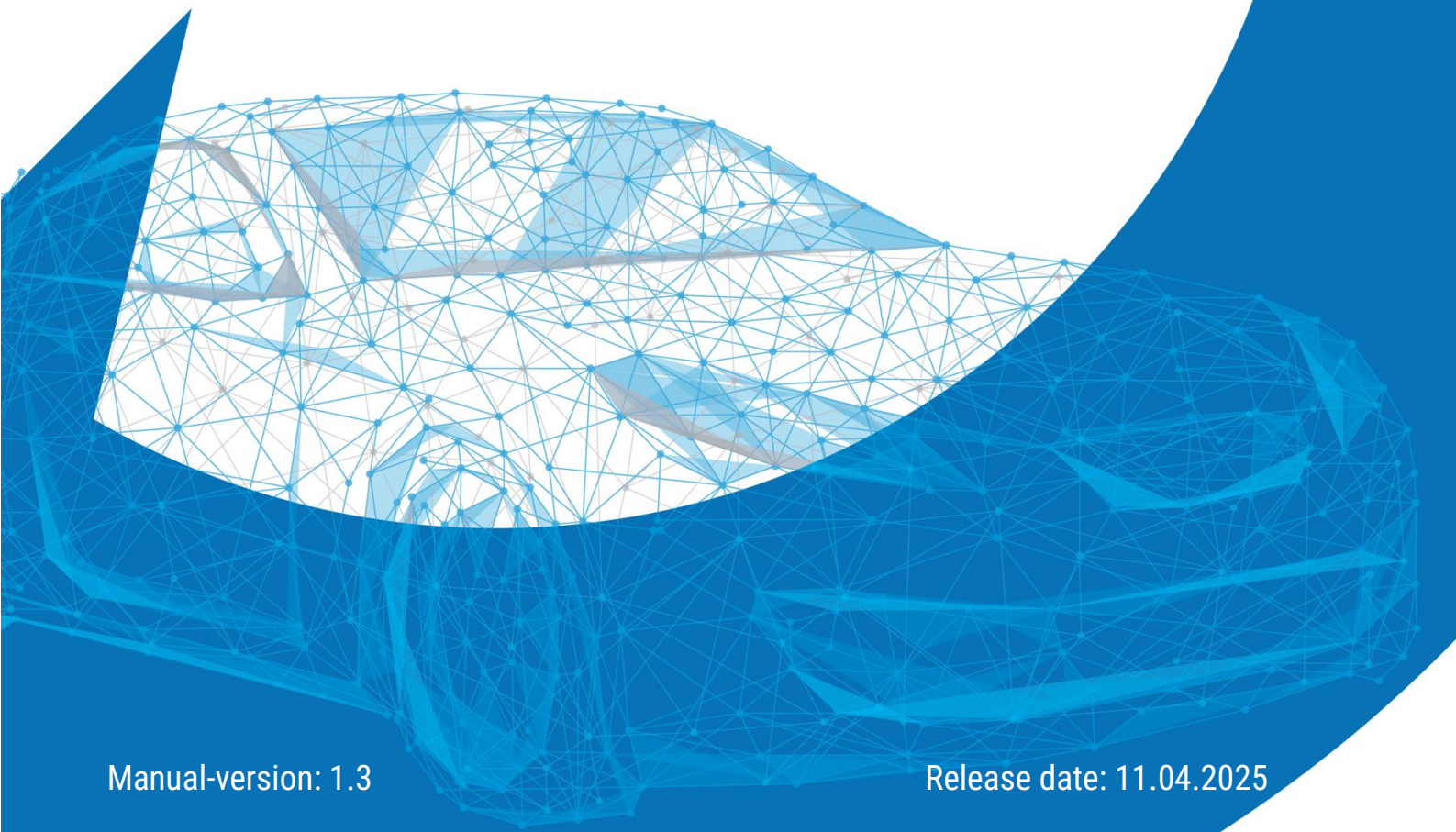
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#OneStepAhead

MediaConverter

100/1000BASE-T1 TC10

User Manual



Manual-version: 1.3

Release date: 11.04.2025

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1 Introduction

This document provides important information and instructions for the reliable, safe, and proper use of Technica Engineering devices. It covers key aspects related to the operation and configuration of the MediaConverter 100/1000BASE-T1 TC10. Please read the User Manual thoroughly before using and adhere to all the information and instructions provided.

If you have any questions or need further clarification, please contact our customer support at support@technica-enginnering.de

1.1 Validity

This user manual is valid for the MediaConverters variants:

- MediaConverter 100/1000BASE-T1 TC10 - MATEnet automotive connector (art. nr. TE-1405)
- MediaConverter 100/1000BASE-T1 TC10 - H-MTD automotive connector (art. nr. TE-1406)

The following hardware versions and software versions of MediaConverters 100/1000BASE-T1 TC10 are valid for this document:

PCBA (HW) version	Product version	Software version
1.5	2.0	R.004.008

Ensure that the User Manual refers to the hardware variant you are using, and the firmware version installed on it.

The User Manual is valid from the time of publishing to the publication of a new version of this User Manual.

Current version in the Customer portal
The firmware is constantly being further developed, and the User Manual is therefore updated accordingly. The latest version of the User Manual can contain different or additional information or instructions. For this reason, always download the latest version of the User Manual from the Technica Engineering´s Customer Portal from https://portal.technica-engineering.de/downloads , before using the device refer to the instructions which are presented in the User Manual.

1.2 Important notes

We provide our important notes and safety instructions in two languages: Englisch (EN) and German (DE). For more details, see the relevant sections:

- EN: Important notes: Important notes
- DE: Wichtige Hinweise: 10 Wichtige Hinweise

1.3 Warranty

We reserve the right to update the contents of this documentation and software without prior notice to reflect improvements or changes. We welcome feedback to enhance our products in the future.

The information in this manual is protected by copyright laws. Unauthorized reproduction or distribution of any part of this manual is prohibited and subject to legal action. For permission or further details, please contact the Technica Engineering directly.

2 Product description

Automotive ethernet, though similar in functionality, has some differences in comparison to standard ethernet. To interact between the two technologies, a converter is needed. The Technica Engineering 100/1000BASE-T1 MediaConverter establish a physical layer conversion between Automotive Ethernet IEEE connections and any device with a standard Ethernet Network Interface Card with a RJ-45 connector with no packets stored or modified.

2.1 Order information

Product Name	Product Article Number	Cable set art. number*
MediaConverter 100/1000BASE-T1 MATEnet	TE-1405	KS-1402
MediaConverter 100/1000BASE-T1 H-MTD	TE-1406	KS-1403

*Cable set needs to be ordered separately

2.2 General description

The Technica Engineering´s MediaConverters are very compact and convenient interface products designed for Automotive Ethernet IEEE 100BASE-T1/1000BASE-T1 and Standard Ethernet IEEE 100BASE-TX/1000BASE-T. These devices are used for the analysis of Ethernet networks, as well as for simulation and testing tasks in Ethernet environments. The supported wakeup/sleep mechanisms over Automotive Ethernet according to the Open Alliance TC10 are used to trigger the startup and shutdown of ECUs in the vehicle network side. This is potentially used for the first startup/wakeup of central ECUs when the car is started by the driver.

MediaConverters are available in two variants with either MATEnet or H-MTD automotive connector.

MATEnet variant



H-MTD variant



Figure 2-1 MediaConverter 100/1000BASE-T1 TC10 in MATEnet and H-MTD variant

2.3 Features

- Conversion of 100/1000BASE-T1 signals in standard 100BASE-TX/1000BASE-T Ethernet signals (bi-directional)
- Status LEDs
- 4x DIP switches for manual configuration
- Console (serial interface) for configuration through command line
- WEB GUI App for user-friendly advanced configuration

- Transceiver Marvell 88Q2221M B2
- Firmware updates
- Frame generator
- Diagnostics

2.4 Technical data

Electrical, thermal and mechanical specifications of MediaConverter 100/1000BASE-T1:

Operating Temperature	-40 °C to +85 °C
Supply Voltage	6 V to 30 V DC (typ. 12 V)
Power connector	MQS 6-pin
Power consumption	2 W
IP Protection Class	IP20
Housing Dimensions	89 mm (W), 72 mm (L), 28 mm (H)
Weight	approx. 0,2 kg
Interfaces	1x 100BASE-T1/1000BASE-T1 (MATEnet/H-MTD) 1x 100BASE-TX/1000BASE-T, RJ45 Micro USB-B

3 Hardware interfaces

In this chapter there is an overview of all HW interfaces of the MediaConverter 100/1000BASE-T1 on the front side and back side.

3.1 Front side

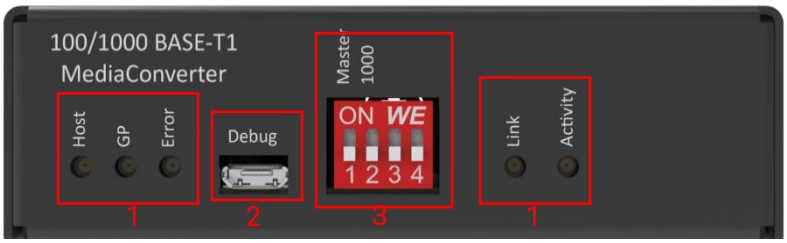


Figure 3-1 Front side of MediaConverter

1. Status LEDs
2. Debug port
3. DIP switches

3.1.1 Status LEDs (1)

The MultiGigabit MediaConverter has 5 LEDs on the front side of the case. LEDs' meaning is described in the table below.

LED	Description
Host LED	Will blink slowly (once a second), during normal operation of the device.
GP LED	General-purpose LED. will be lit when the PHYs are configured. If the PHYs fail to get configured this LED will be off.
Error LED (red)	Will be lit when at least one CRC error is detected.
Link LED	BASE-T1 Link active.
Activity LED	BASE-T1 Ethernet data transmission.

Table 3-1 Status LED´s meaning description

If the GP LED is ON and Host LED is off, this indicates an error state. In case auto search feature is enabled GP LED will blink quickly (every 100ms). In case automation command is enabled, the GP LED will follow a special blinking pattern.

If the error LED is lit red, a disconnection of the link can clear the LED. This can be done by changing the DIP switch position to break the link and changing it back to form a link again. To reset the CRC error counter is necessary to provide power reset of device.

3.1.2 Debug port (2)

The debug micro-USB connector can be used as a serial interface to interact with the device.

A micro-USB-B cable is needed for connection to the service connector. Cable is not part of the delivery and cannot be ordered from Technica Engineering.
This interface enables the user to read TX/RX register counters, SQI values of the channels, CRC errors, and other information. It can also be used for bootloader updates of the device. For more info, see chapter 4.3.

3.1.3 DIP Switches (3)

DIP switches on the front side of the device are used for manual configuration of the device. You can find more information about configuration in chapter 4.

3.2 Back side

On the picture below are shown two variants of the MediaConverter TC10 with H-MTD or MATEnet automotive ethernet connectors.



Figure 3-2 Back side of MediaConverters

4. Standard ethernet connector
5. Automotive ethernet connector (MATE
6. MQS 6 pin connector

3.2.1 Standard ethernet connector (4)

This is a standard RJ-45 connector to be used with CAT6 cable for 100BASE-TX or 1000BASE-T connections.

3.2.2 Automotive ethernet connector (5)

The device is equipped with standard ethernet connectors: MATEnet (TE-1405 variant) or H-MTD (TE-1406 variant). In case of purchasing official cable sets from Technica Engineering, the connectors are part of the cables.

MATEnet connector

Name	Picture	Part Number
1-PORT MATENET FRAME, UTP COD Z		2302454-9
Tyco MATENET, FEMALE INSERT MODUL, BLACK		2302450-1
Tyco Nano MQS crimp contact		1-1703930-2

Pinning:

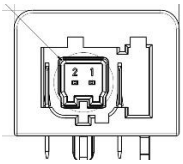


Figure 3-3: MATEnet Connector

Pin	Function
1	Data Line Minus (Negative)
2	Data Line Plus (Positive)

Table 3-2 MATEnet connector pinning

H-MTD connector

MediaConverter TE-1406 is equipped with a standard shielded H-MTD connector. The shield in the device is connected to the ground by multiple 10nF and 100nF capacitors. Also, by one 100Kohm resistor and a varistor Diode.

Name	Picture	Part Number
Rosenberger H-MTD Stecker		E6K10A-1CAZ5-Z

Table 3-3 Rosenberger H-MTD Connector

Pinning:



Pin	Function
1	Data Line Minus (Negative)
2	Data Line Plus (Positive)

Table 3-4 H-MTD connector pinning

3.2.3 MQS connector (6)

MediaConverter 100BASE-T1 has from the back side of the device MQS 6 pin-connector. This connector is used for power and for automotive connections.

Pinning:

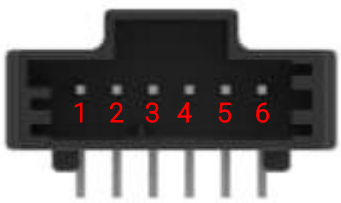


Figure 3-4 MQS pin numbering

Pin	Function
1	Reference GND only
2	Link Status Output (3V3, Active Low) (LS)
3	Force Link Down Input (Active Low) (LD)
4	Force Slave Mode (Active Low) (FS)
5	Power GND
6	VBAT (6 to 30 Volt DC)

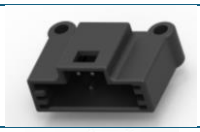
Name	Picture	Part Number
TE Connectivity 6pos MQS .63 header 90deg THR		1-1418888-5
TE Connectivity MQS BU-GEH 6P		1-969508-2 Distributor: mouser.de

Table 3-5 MQS connector description

MQS crimp contacts are already crimped on the cables, when purchased official Cable set from Technica Engineering. If needed to crimp Tyco contacts, then Technica Engineering recommends using official crimp tool – CS11K NANO-MQS, 0,13-0,35 SQ.M

4 Configuration

The MediaConverters from Technica Engineering can be configured in multiple ways.

4.1 Manual configuration

The MediaConverter 100/1000BASE-T1 can be configured by four **DIP switches** on the front side of the device.

DIP Switch	Status	Description
1	ON (up) OFF (down)	BASE-T1 Port is set to Master BASE-T1 Port is set to Slave
2	ON (up) OFF (down)	1000 BASE-T1 to 1000 BASE-T conversion 100 BASE-T1 to 100 BASE-TX conversion
3	ON (up) OFF (down)	Auto Search Enabled Auto Search Disabled
4	ON (up) OFF (down)	Frame Generator active on 100/1000BASE-T1 Normal operation

Figure 4-1 DIP switches configuration

For establishing a BASE-T1 link one device must be set as Master and the other must be set as Slave.

It is not possible to mix the two speeds of the ports. If the MediaConverter is configured for 100BASE-T1, the RJ-45 connector is always 100BASE-TX. If the MediaConverter is configured for 1000BASE-T1 the RJ-45 connector is always 1000BASE-T.

The Auto Search Feature is designed to dynamically adjust the configuration of the BASE-T1 port to establish a successful link. This process involves systematically changing the port's operating mode (MASTER/SLAVE) and speed (100/1000 MBPS) until a link up is detected. Otherwise, it will keep on checking all possible configurations.

4.2 Configuration through power MQS connector

The PINs 2, 3, and 4 of the Power MQS connector can be used for configuration and link status output.

4.2.1 Reference Pin (Ref) (PIN 1)

Pin 1 is intended only for ForceSlave, ForceLinkdown, and LinkStatus reference ground connection. If pin 2, 3, and 4 are not connected then pin 1 does not have to be connected.

4.2.1 Link status output (LS) (PIN 2)

This PIN is active low. When there is a link up on BASE-T1 port the voltage on PIN 2 is 0 volt. The link status PIN 2 outputs a weak 3.3 Volts status signal in case of no linkup. There is an integrated 5K Ohm pull-up resistor to 3.3 Volts. The pull-down transistor can drain up to 100mA.



Figure 4-2 Link status output

4.2.2 Force link down (LD) (PIN 3)

If you connect PIN 3 and PIN 1 by a shortcut wire link down is forced. These pins are active low. There is an integrated 5K Ohm Pullup to 3.3 Volts. For normal operation please disconnect the PINs.

While removing the shortcut of PIN 3 and PIN 1, a full reset of the device will be done.

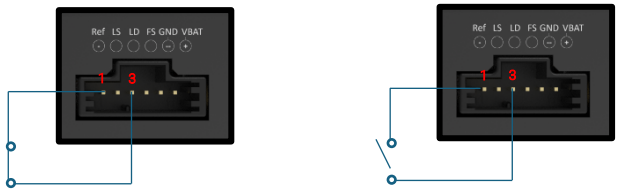


Figure 4-3 Force link down

4.2.1 Force slave mode (FS) (PIN 4)

If you connect PIN 4 and PIN 1 by a shortcut wire, Force Slave mode is activated. These pins are active low. There is an integrated 5K Ohm Pullup to 3.3 Volts. For normal operation please disconnect the PINs.

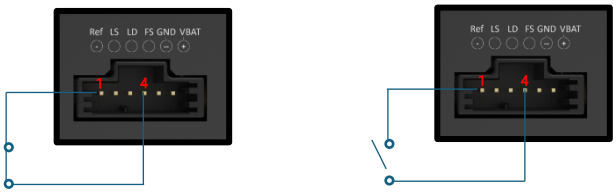


Figure 4-4 Force slave mode

The activated Force Slave mode overrides the DIP switch configuration.

Power reset is not needed

4.3 Configuration/Automation options through USB-serial console interface

The debug micro-USB connector can be used as a serial interface to interact with the device and dynamically reconfigure it, making very suitable by any remote automation use case. User has two options for advanced configuration via serial Interface. In this chapter configuration possibilities via commands (console) using any terminal application, like TeraTerm, are described. More user-friendly configuration can be done via MediaConverter Web GUI PC application from Technica Engineering, which is described in chapter: 5.

To start communication between PC and MediaConverter is necessary to connect micro-USB-B – USB-A cable to debug connector of MediaConverter and USB-A connector to your PC. The MediaConverter will automatically appear as a virtual COM port on the USB. Media Converter should be automatically

recognized in the host computer, for both Windows and Linux OS by using USB CDC (Communication Device Class) drivers.

To successfully start serial communication between the PC and MediaConverter, it is necessary to set up the terminal application (e.g.: Tera Term) according to the following instructions:

1. Connect the debug connector (4) to your PC and to the device.
2. Open Teraterm or any other serial port terminal application and select the corresponding Serial COM port.

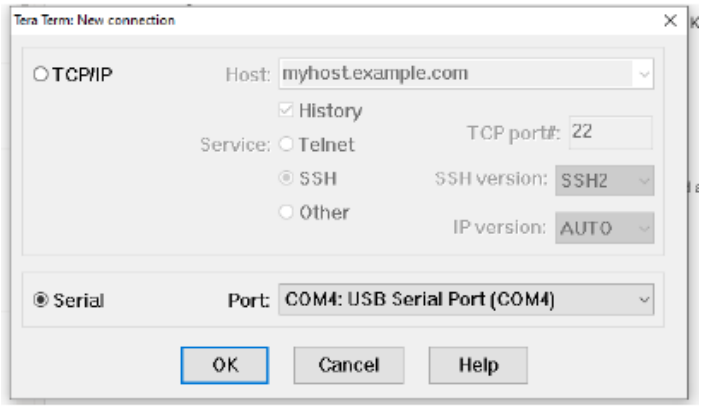


Figure 4-5 TeraTerm new connection window

3. Configure serial port settings. The speed must be configured to 115200bps-8N1 (1 start bit, 8 data bits, no parity bit, 1 stop bit, no flow control).

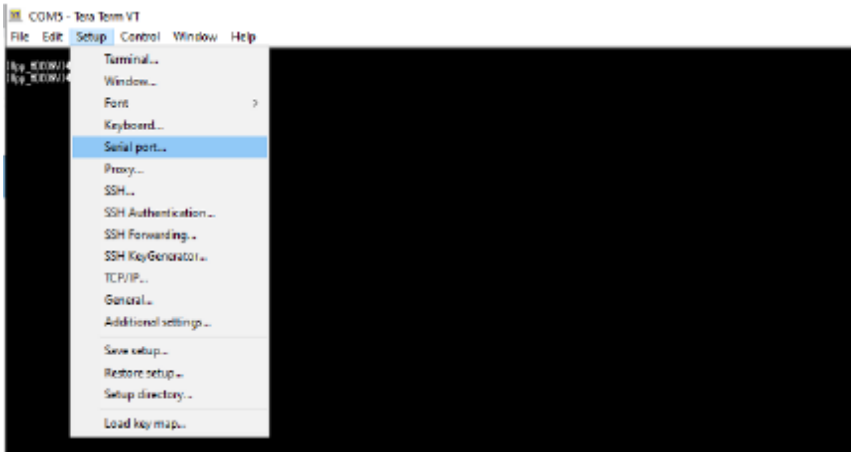


Figure 4-6 Setup tab

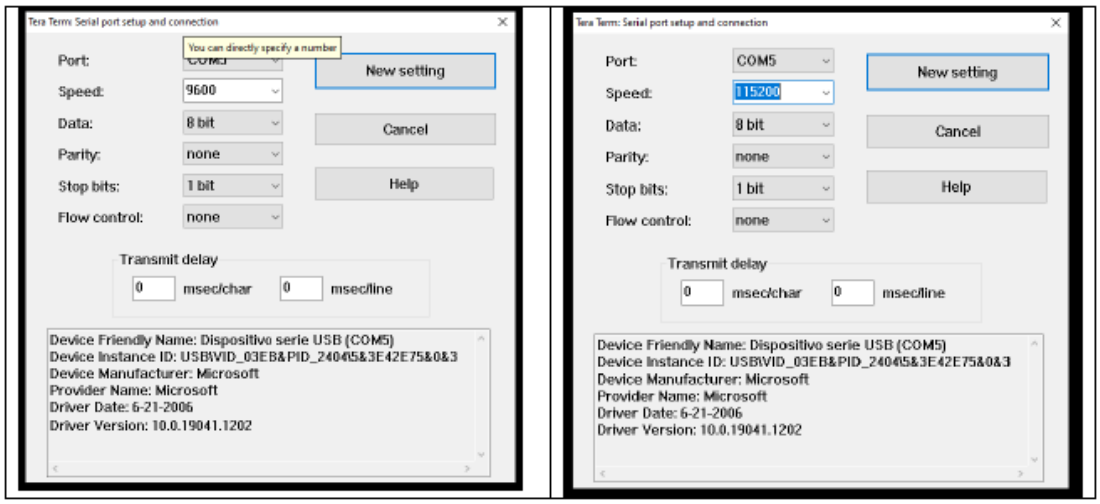


Figure 4-7 Serial port setup and connection

4. Press enter to validate that access to the device is confirmed. It will appear something like:

To view the commands for the Debug connector, type "help". This will list all the possible options available for the parameters listed in the MediaConverter´s console.

```
List of commands
            Command : Description
-------------------------------------------------------------------
               help : Prints a list of commands
               echo : Control console echo mode
              clear : Clears screen
                log : Changes logging level
               mdio : MDIO diagnostic commands
             macsec : MACsec configuration commands
             status : Show status info of the current board
                fbl : Jumps to bootloader
              reset : issues reset
            version : Show system information
               auto : Automation - Set up a custom device configuration
               tc10 : TC10 - Manage Marvell TC10 own implementation commands
           testmode : Switch to test mode
                phy : PHY related commands
            license : Show license status
[App_MDCONV]#>
```

Figure 4-8 List of commands

Command echo:
This command enables/disables the visualization of characters on the screen.

Syntax:
- echo: displays the current state of character display, ON or OFF.
- echo on: enables the display of each character in the console. Every time a command or text is entered, the characters will be displayed on the screen.
- echo off: disables the display of characters in the console. When a command or text is entered, the characters will not be displayed on the screen.

Usage:
- To determine whether character display is currently enabled or disabled, enter the echo command with no parameters.
- To enable character display in the console, enter the echo on command.
- To disable character display in the console, enter the echo off command.

Command log:

Log command lets to select a logging level from 1-5, where the higher logging level the more information is printed.

Syntax:
- **log**: returns the current logging level.
- **log** [level]: changes the logging level to the specified [level], where the level must be a number between 1 and 5 (inclusive).

Usage:
- To view the current logging level, enter "log".
- To change the logging level, enter "log" followed by the desired [level] between 1 and 5.
- To view the syntax, enter "log" followed by any character different from the range 1 to 5.

Command mdio:

Mdio command allows to read and/or write operations for clause22, clause45 for each variant.

Syntax:
- **mdio**: displays the mdio submenu syntax.
- **mdio help**: displays help information.
- **mdio list**: shows the list of MDIO buses.
- **mdio read22** [bus] [phy_addr] [reg_addr]
- **mdio write22** [bus] [phy_addr] [reg_addr] [value]
- **mdio read45** [bus] [phy_addr] [dev_addr] [reg_addr]
- **mdio write45** [bus] [phy_addr] [dev_addr] [reg_addr] [value]

Usage:
- To deploy the mdio submenu, enter "**mdio**" or "**mdio help**".
- To show the list of mdio buses, enter "**mdio list**".
- To read a specific register clause 22, enter "**mdio read22 [bus] [phy_addr] [reg_addr]**".
- To write a specific register clause 22, enter "**mdio write22 [bus] [phy_addr] [reg_addr] [value]**".
- To read a specific register clause 45, enter "**mdio read45 [bus] [phy_addr] [dev_addr] [reg_addr]**".
- To write a specific register clause 45, enter "**mdio write45 [bus] [phy_addr] [dev_addr] [reg_addr] [value]**".

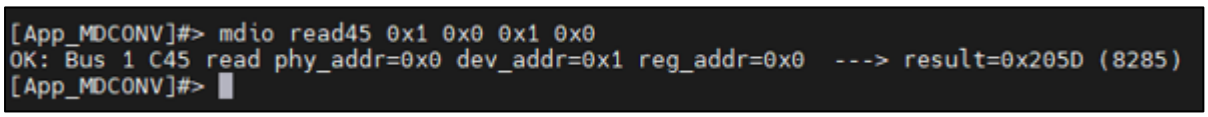


Figure 4-9 Example of media command

Command status:
The status prints the Phy parameters status: DIP switches, Mode (Master/Slave), Speed, Legacy mode, Base-T1 status (Link, speed, polarity and quality) and SerDes status.

Syntax:
- **status**: displays the mdio submenu syntax.
- **status help**: displays help information.
- **status run**: the run returns an image of status.
- **status poll**: the poll keeps the status screen and refreshes it every 500ms.

Usage:
- To deploy the status submenu, enter "**status**" or "**status help**".
- To show an image of the status, enter "**status run**"
- To show continuous the status, enter "**status poll**"

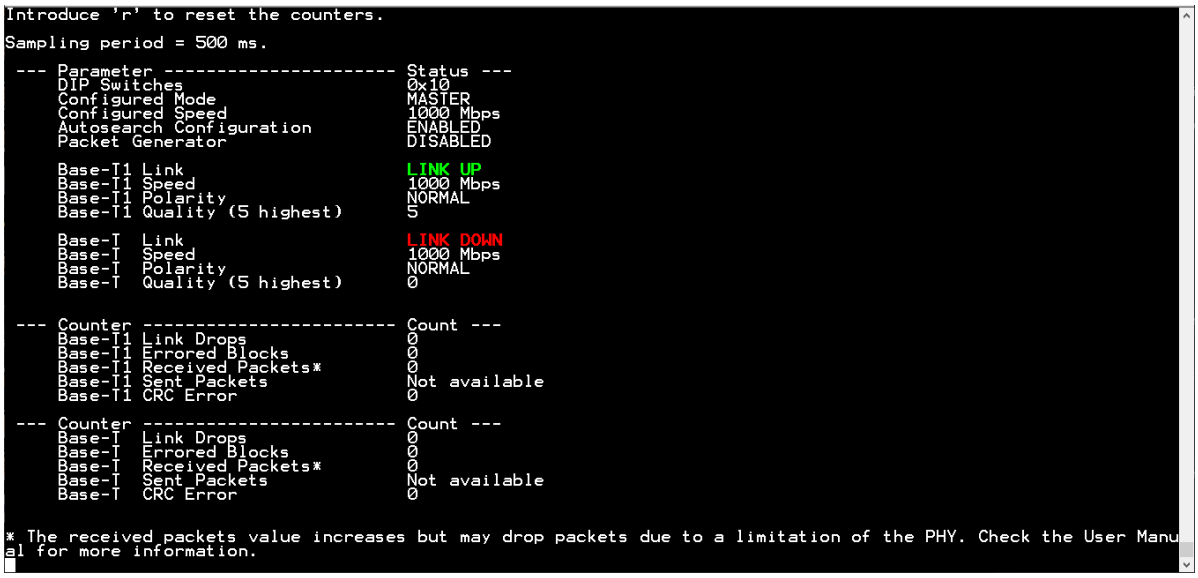


Figure 4-10 Status run command example

Command fbl:
Jumps to Bootloader, see chapter 7.

Command reset:
Resets the full equipment.

Command version:
Prints information regarding the product: System release, App release name, date, app hash, HW variant, version and connector

Command auto:
The auto command has two states: ON or OFF. When OFF, retrieves the Phy configuration from the DIP Switches position, when ON allows the configuration to change.

Syntax:
- auto help: Prints the auto submenu.
- auto set: mode [master/slave] speed [100m/1000m] search [on/off] gen [on/off]
- auto off: Disables auto command.
- auto get: if auto is ON, retrieves the automation configuration. Otherwise, it will return an Error message: "*Error: Automation mode is not active. There is no active automation config.*"
- auto link: retrieves the link status (link up or link down)

**Note: When Packet Generator is enabled via automation command (gen on), the default configuration is used. See* Table 42

Usage:
To overwrite the configuration of the DIP switches via the console (master/slave, speed, auto search, packet generator). This configuration remains set until the MC is restarted. The parameters of the auto command can be used independently. In other words, the desired parameters can be modified without affecting others.

Example:

The current configuration is set as it is on Figure 49. If a user wants to change this configuration without touching DIP switches, is possible to set it in console as follows:

After typing command auto, will be listed possibilities of auto command:

```
[App_MDCONV]#> auto

Automation mode selection syntax:

 auto help   : This message
 auto set    : Sets a automation config and resets the PHY when the new config is valid.
 auto off    : Disables the automation mode and resets the PHY. The DIP switches are responsible for the config.
 auto get    : Pulls current device automation configuration.
 auto link   : Retrieves Base-T1 link status
```

Figure 4-11 Auto commands

To override configuration from DIP switches, is necessary to follow syntax: "auto set mode [master/slave] speed [10g/5g/2.5g]". For example, to set MASTER mode, 5G speed:

```
[App_MDCONV]#> auto set mode master speed 5g
OK: Configuration applied
[App_MDCONV]#>
```

Figure 4-12 Example of auto configuration

To enabled auto search feature:

```
[App_MDCONV]#> auto set search on
OK: Configuration applied
[App_MDCONV]#>
```

Figure 4-13 Example of auto configuration

After successful application, is possible to check if configuration was applied either with the "auto get",

```
[App_MDCONV]#> auto get
mode: master
speed: 5 Gbps
search: on
gen: off
```

Figure 4-14 Example of auto get command

or with command "status run/poll" *command.*

Command PHY:
The phy command allows additional actions or configurations of the phy.

Syntax:
- phy help: Prints the phy submenu.
- phy mute: performs the TX_DISABLE to mute the PHY.
- phy unmute: release the TX_DISABLE to unmute the PHY.
- phy status: Prints if phy is muted or unmuted.
- phy reset [hard/soft]: Reset the PHY (hard / soft).

Command testmode:

Testmode command allows to set and select a specific test mode.
Syntax:
- **testmode help:** Prints the testmode submenu.
- **testmode get:** Returns the current mode selected. By default, test mode 0 - Jitter pattern 0.
- **testmode set:** Sets the desired test mode or jitter pattern
- **testmode off:** Resets the test mode and jitter to normal state

Usage:
- To deploy the test mode submenu, enter "**testmode**" or "**testmode help**".
- To get the current test mode selected, enter "**testmode get**".
- To set a new mode, enter "**testmode set [mode]**" (with mode from 0 to 7]) with one of the listed modes.
- To reset any test mode or jitter pattern previously selected, just enter "**testmode off**" and it will be configured as default test mode 0, which corresponds to Normal Operation.

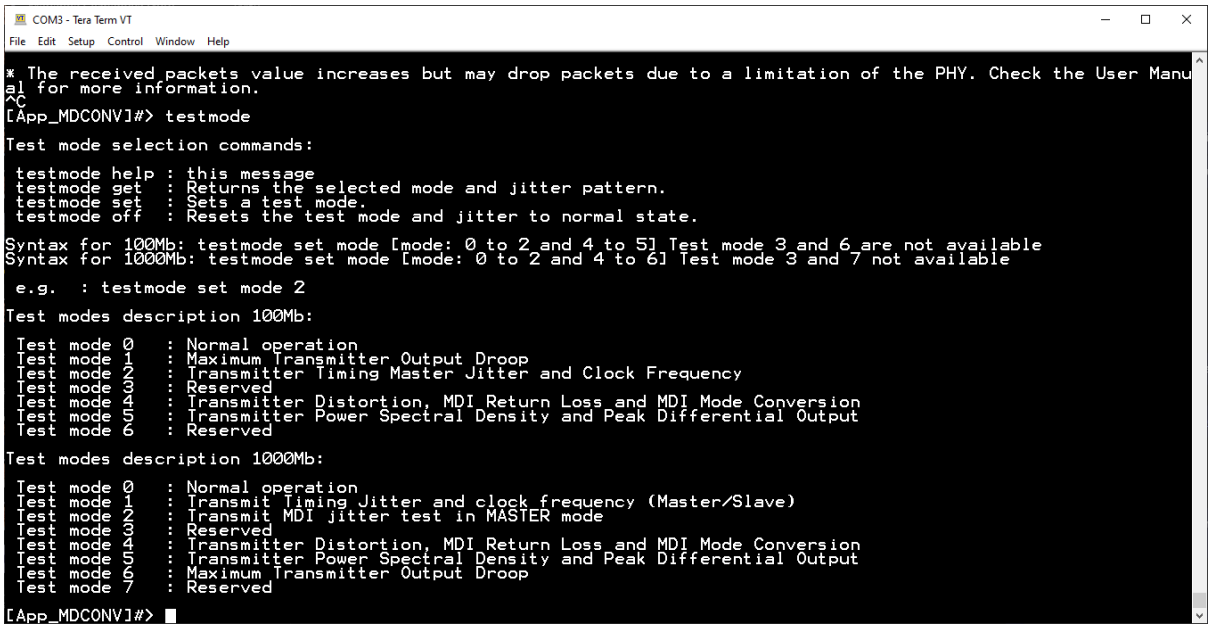


Figure 4-15 Testmode command output with explanation

Example:

```
[App_MDCONV]#> testmode set mode 5

Test mode 5: Transmitter Power Spectral Density and Peak Differential Output

Set test mode 5 successfully

[App_MDCONV]#>
```

Figure 4-16 Testmode 5 example

5 GUI-Based configuration (PC application)

A dedicated PC application provides a graphical user interface (GUI) like a web-based interface but runs locally on a computer. This method offers an intuitive way to configure and monitor the device. Installation and usage details can be found in the next chapters.

5.1 System requirements

The application is available for Windows 11 and Ubuntu 22.04. The WEB interface is not supported in the latest versions of Ubuntu (i.e. Ubuntu 24.04).

5.2 Installation

5.2.1 Linux-based

The GUI-Based PC application can be found on Technica Engineering Customer Portal on link: https://portal.technica-engineering.de/downloads

To install the application, run:

sudo dpkg -i Media_Converter_App_MC_WEB_XXX.XXX.XXX.deb

or

sudo apt install ./Media_Converter_App_MC_WEB_XXX.XXX.XXX.deb

from the directory that contains the file .deb.

5.2.2 Windows

To install the application, execute:

Media_Converter_App_MC_WEB_XXX.XXX.XXX_x86-setup.exe

Follow the next steps to complete the installation:

1. The following windows will appear. Click *Next*.

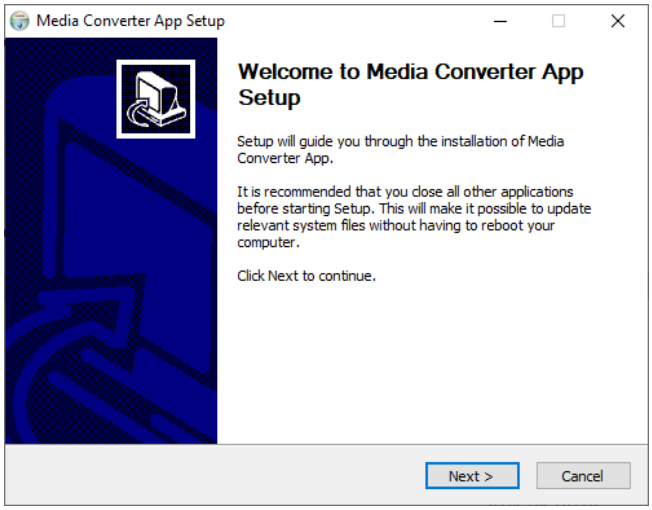


Figure 5-1 Installation guide Step 1

2. Choose the directory of the destination folder to install the Media Converter App or leave the one by default. Then, click **Next**

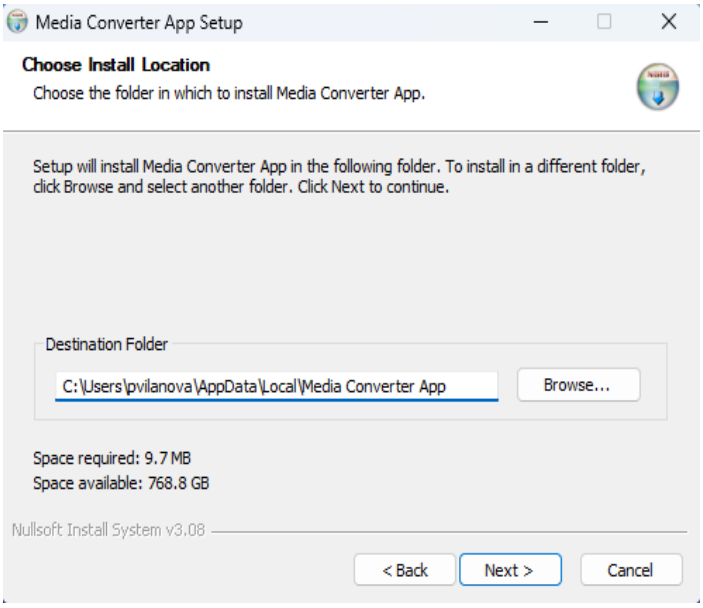


Figure 5-2 Installation guide Step 2

3. Click Install.

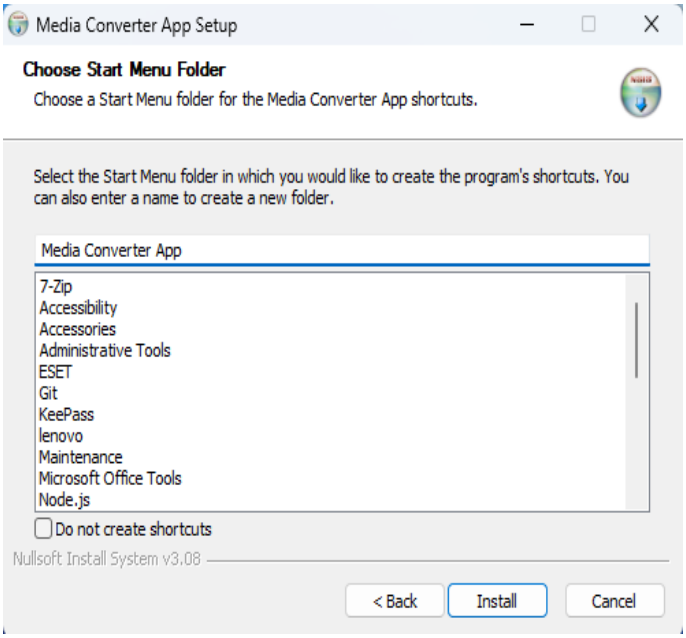


Figure 5-3 Installation guide Step 3

4. Wait until the installation is completed and click *Next*.

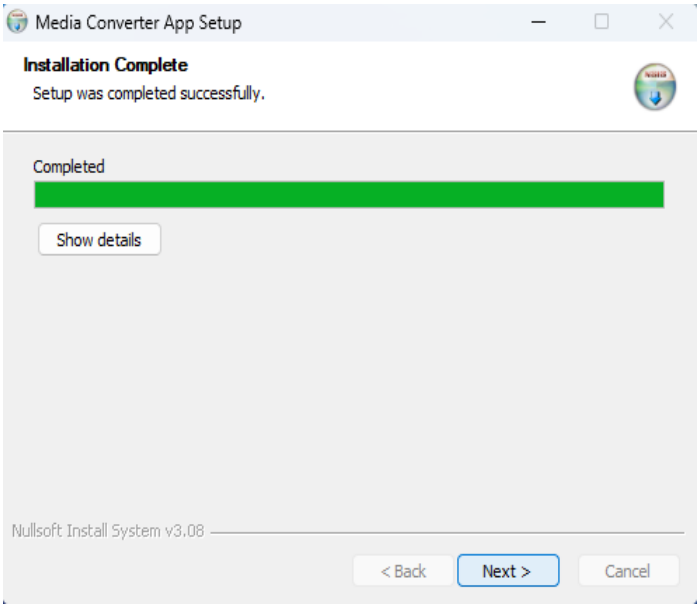


Figure 5-4 Installation guide Step 4

5. Click *Finish*.

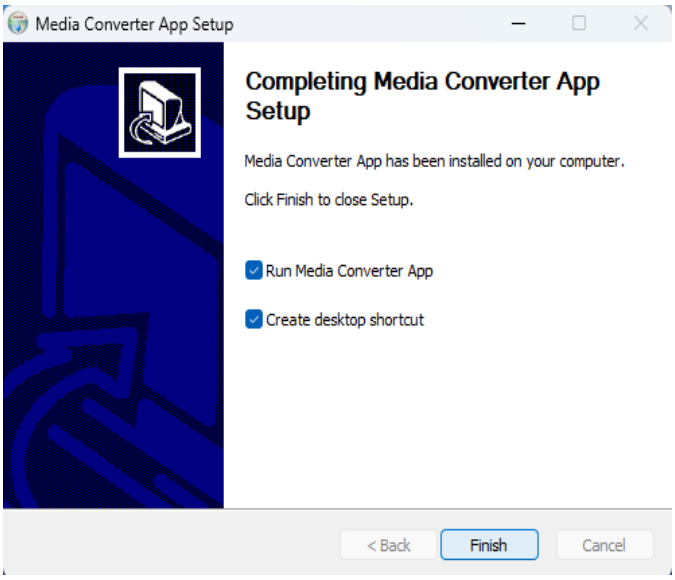


Figure 5-5 Installation guide Step 5

The application will run automatically if the option Run Media Converter App is enabled. Disable this option if it is desired to open the application later.

5.3 Connecting the device

Once the application is opened, the default window is shown:

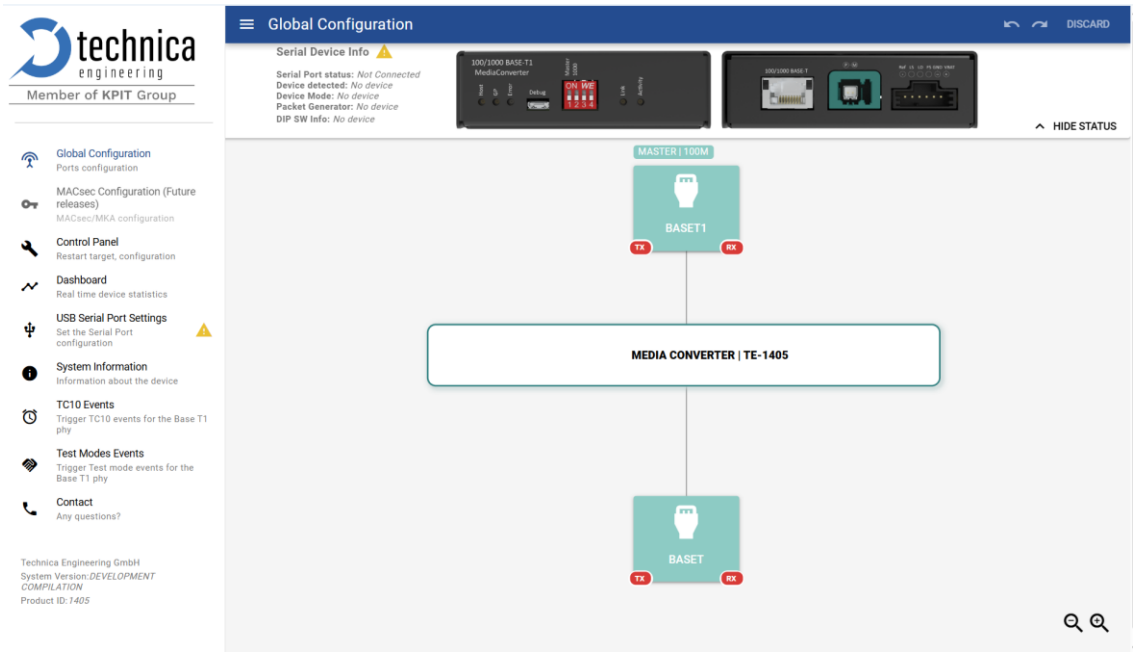


Figure 5-6 Default Web Interface

When the application is opened, the default configuration is used, and it will be adapted once a MediaConverter is chosen.

As can be seen, a warning icon is displayed. It indicates that no MediaConverter is connected to the application. Therefore, go to USB Serial Port Settings panel by clicking in the left menu tab:

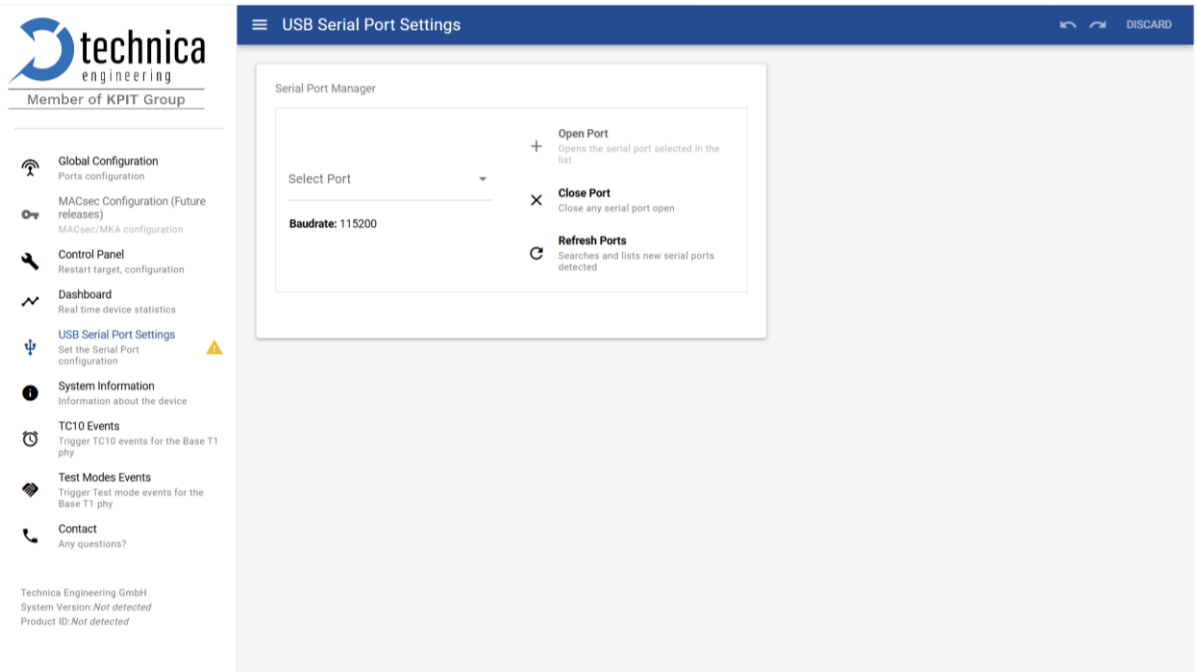


Figure 5-7 USB Serial Port Settings Panel

This panel will manage the serial port interfaces connected to our PC. Three options are available:

- *Open Port*: once the USB port is selected, the application will open it
- *Close Port*: the application closes the USB port if opened
- *Refresh ports*: the application searches within all the ports connected to the computer. All available ports will be shown in the Select Port component by clicking the down arrow icon.

Select the desired port from the *Select Port* menu and click *Open*. The application will automatically detect the connected Media Converter and adjust the GUI based on the product ID. For TE1405:

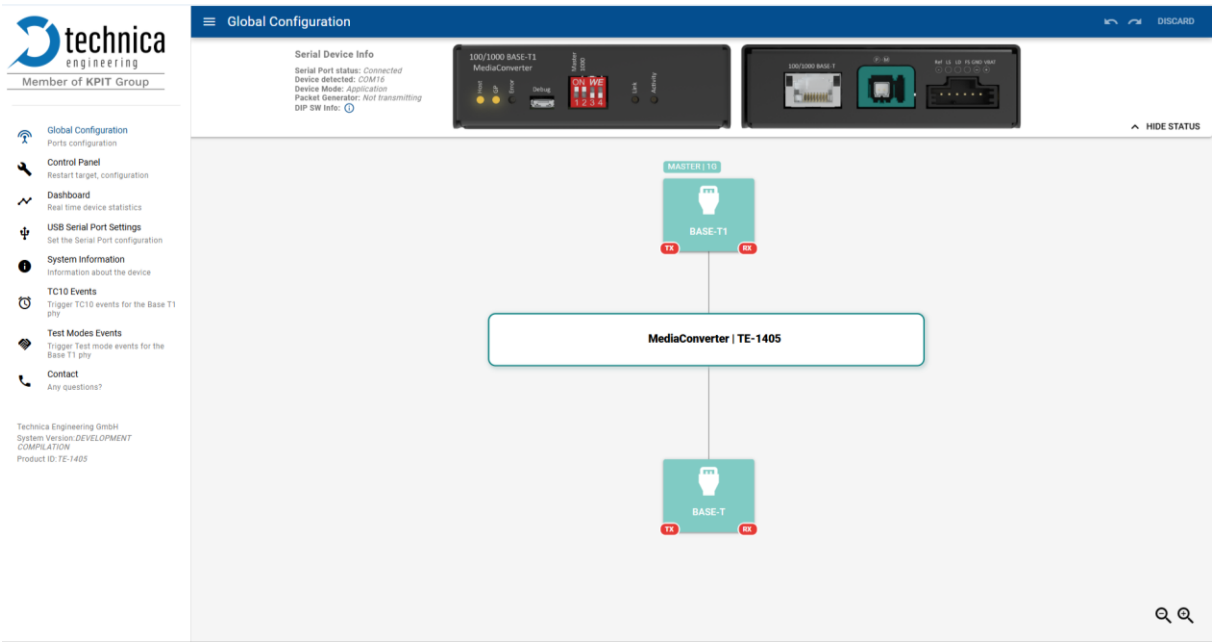


Figure 5-8 Global Configuration panel

5.3.1 Devices supported

The WEB GUI interface supports the following combinations of hardware and software:

Product ID	Product Version	System SW Release
TE-1406	1.3	R.004.008
TE-1406	1.4	R.004.008
TE-1406	1.5	R.004.008
TE-1406	2.0	R.004.008
TE-1405	1.3	R.004.008
TE-1405	1.4	R.004.008
TE-1405	1.5	R.004.008
TE-1405	2.0	R.004.008

Table 5-1 Variants supported

In case you have an old software release not supported, by the WEB GUI, refer to *7.2.2* in order to use same WEB GUI to perform a software update.t.

5.4 Application overview

This section describes in detail all the features available on this application in each panel from the menu.

5.4.1 Global Configuration

In this panel, the status of both ports configuration is shown BASE-T1 port and BASE-T port. Depending on the link status of the port, the BASE-T1 icon will be the following:

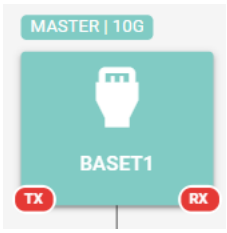


Figure 5-10 Link down BASE-T1 port

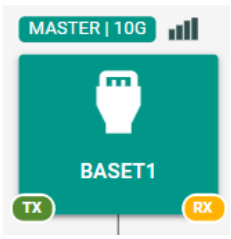


Figure 5-9 Link up BASE-T1 port

and the BASE-T:

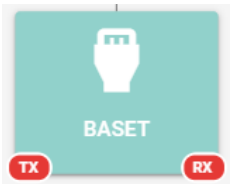


Figure 5-11 Link down BASE-T1 port

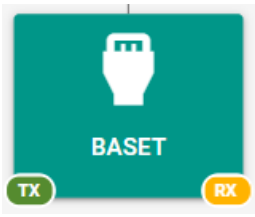


Figure 5-12 Link up BASE-T1 port

If the user places the mouse in the port icon, the following hovers will show:

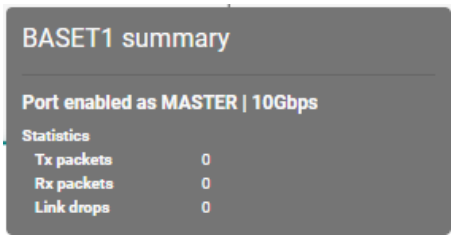


Figure 5-14 BASE-T1 hover information

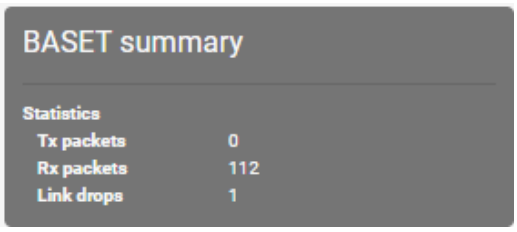


Figure 5-13 BASE-T hover information

In case the auto-search feature is enabled, the port icon behavior will be the following:

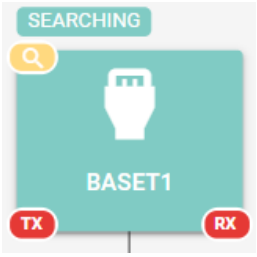


Figure 5-15 Searching, link down

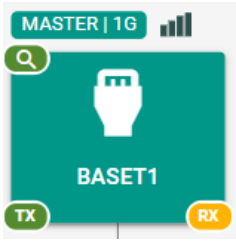


Figure 5-16 Configuration found, link up

5.4.2 Port dialog

- General Configuration

The BASE-T1 also includes a port dialog for additional configuration. The dialog is shown when the BASE-T1 port icon is clicked:

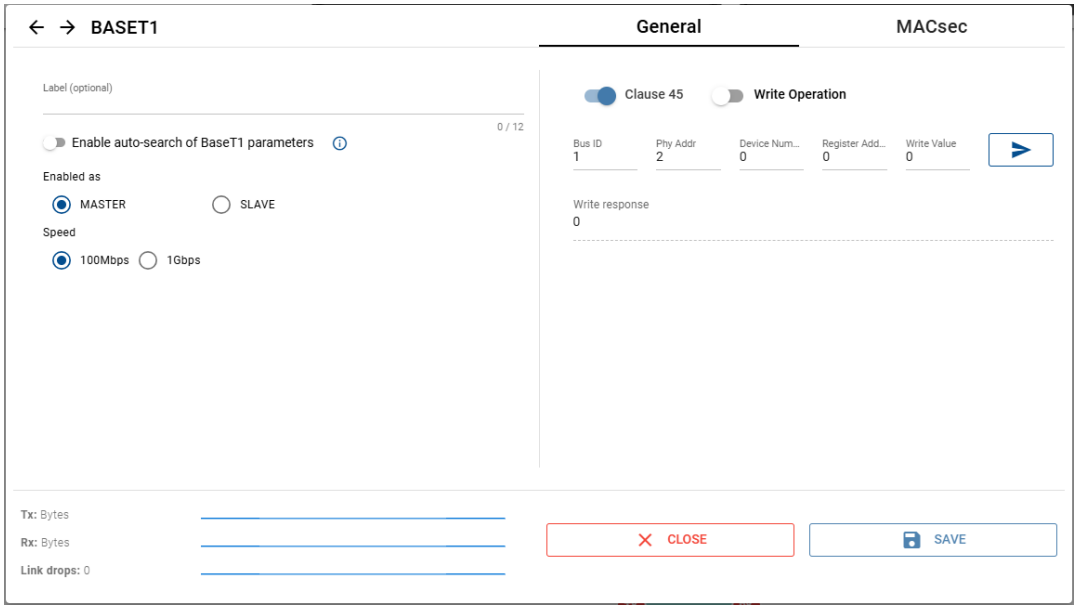


Figure 5-17 BASE-T1 port dialog (General Tab)

The general view includes the following features:

- *Label*: in case a custom name wants to be configured
- *Auto-Search Enable Toggle ("Enable auto-detection of BASET1 parameters")*: This toggle will enabled/disabled the automatic search feature (See Section 3.1.3)
- *Basic Configuration:* 5 buttons are available to set the configuration mode and the speed. This configuration will override the DIP switches configuration once applied.
- *MDIO Interface*: This will ease the MDIO read and write transactions. In this case, BASE-T1 port supports only Clause 45, hence it will remain fixed. Choose whether reading or writing operation is desired and complete the register fields. Then, click blue arrow button to perform the operation. Automatically, the response will be shown.

- MACsec Configuration

As mentioned in Chapter 5.6 MACsec feature is available under license. The web will show if MediaConverter has license in the System Information panel as shown on for available license and Fig.14(b) for license not available.

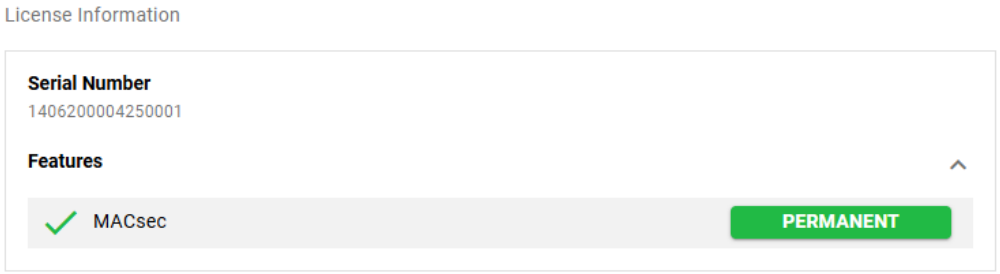


Figure 5-18 Available license

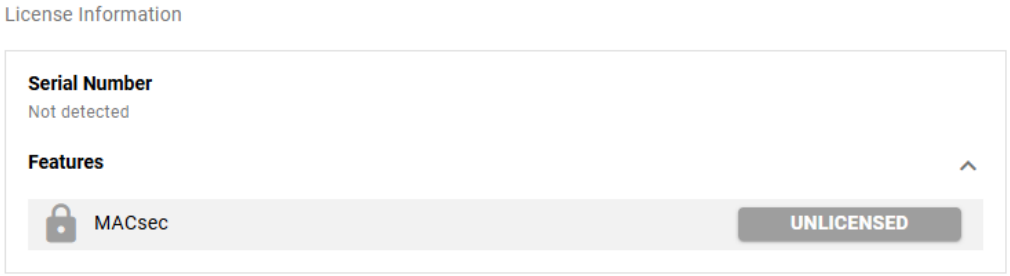


Figure 5-19 Not available license

In case, there is no license available, the "MACsec" tab in the port dialog from the BASE-T1 port will be disabled and if tried to access, a "Unlicensed" tooltip appears to indicate so, as shown here:



Figure 5-20 MACsec feature disabled

In case a MACsec license is available, the following interface will show when “MACsec” tab is clicked.

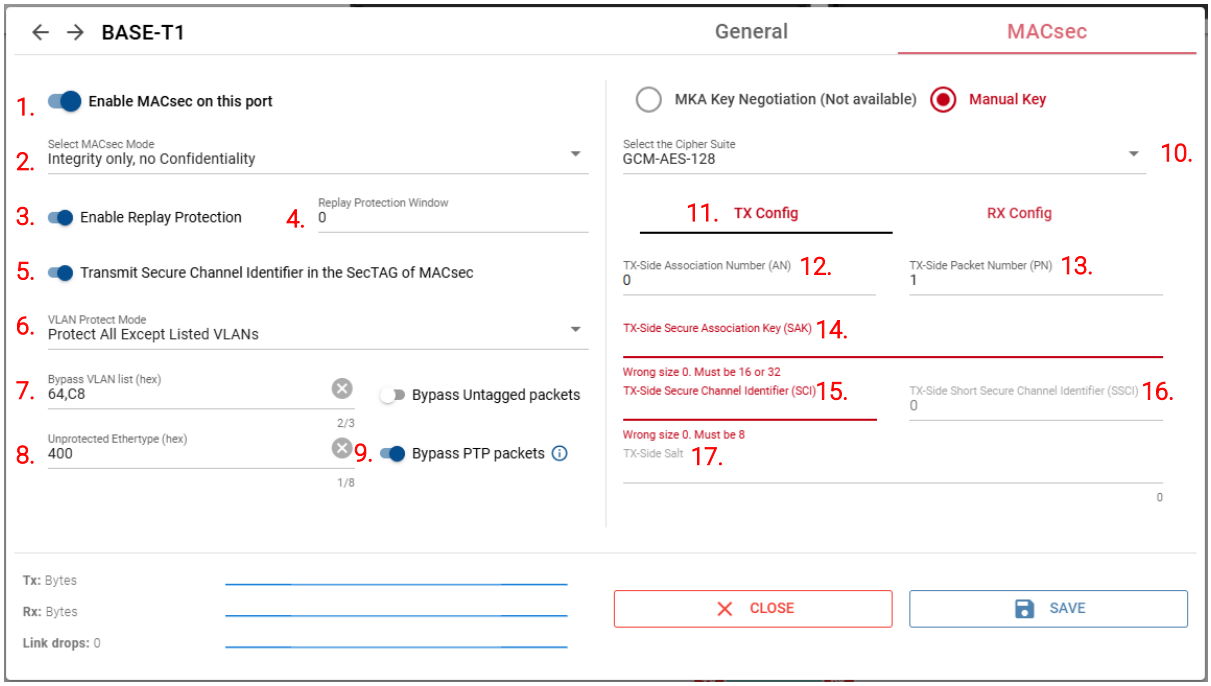


Figure 5-21 MACsec tab on BASE-T1 port

Description of MACsec configuration options:

1. Enable MACsec on this port: This option enables/disables the MACsec functionality on this port. By default, the option is disabled.
2. Select MACsec mode: The user can select possible settings for MACsec mode, from the below selections.

- Neither integrity nor Confidentiality: MKA negotiation without MACSEC
- Integrity only, no Confidentiality: The payload is protected with an integrity signature, but the data is not encrypted
- Integrity and Confidentiality, offset 30: The payload is protected with an integrity signature and all the payload is encrypted except the first 30 bytes
- Integrity and Confidentiality, offset 50: The payload is protected with an integrity signature and all the payload is encrypted except the first 50 bytes

By default, the value is set to “integrity only, no confidentiality”.

3. Enable Replay Protection: This option enables/disables replay protection functionality on this port. By default, the option is enabled.
4. Replay protection window: This field enables the user to specify the replay protection window size. It’s only applied if enable replay protection is active. By default, the value is 0. Possible values are all uint32 values.
5. Transmit secure channel identifier in the SecTAG of MACsec: This option implies that the MACsec tag will contain the MAC and port of the entity with whom it negotiated. Given that this is a switch, the default value is set to enable.
6. VLAN protect mode: This option sets which VLANs need to be protected. The user can select one of the two options:
   - Protected all except listed VLANs
   - Protect only listed VLANs

By default, the option Protect all except listed VLANs is selected.

7. Bypass VLAN/Unprotected VLAN(hex): This field gives a list of VLANs to protect or bypass. By default the field is empty. Possible values are from 1 to 4095. Up to 3 values can be added in protect mode while 2 values can be added in Bypass mode.
8. Unprotected EtherType(hex): This field gives a list of ethertypes that are not going to be encrypted (up to 7 ethertypes can be added to this list). By default, the field is empty.
9. Bypass PTP packets: If this field is enabled and PTP is configured for this port, PTP messages will not be protected by MACsec. This field is enabled by default.
10. Select MACsec ciphers: This option provides a list of ciphers in order of preference. A non-listed cipher is not permitted. Possible ciphers values are:
    - GCM_AES_128
    - GCM_AES_256
    - GCM_AES_XPN_128
    - GCM_AES_XPN_256

11. TX Config: There user must configure the TX parameters here.
12. TX-Side Association Number (AN): The user must input the association number. By default, the value is 0. Possible values are between 0 to 3.
13. TX-Side Packet Number (PN): The user must input the packet number used by the cipher. By default, the value is 1. Possible values are all uint64_t values except 0 if XPN cipher is used, otherwise, all 32bit values are used except 0.
14. TX-Side Secure Association Key (SAK): The user must provide the SAK value. The possible values are between 0, 16 bytes, or 32 bytes. Any size other than these values will result in an error.
15. TX-Side Secure Channel Identifier (SCI): The user must provide the value of Secure Channel Identifier value. The possible values are between 0 and 8 bytes. Any size other than these values will result in an error.
16. TX-Side Short Secure Channel Identifier (SCCI): This field stores an unsigned integer of 32 bits. This can only be used in XPN ciphers. By default, the value is 0. There is no restriction on possible values.
17. TX-Side Salt: This field stores 12 bytes as a value for the Salt used in XPN ciphers. By default, the value is kept empty. Possible values are between 0 and 12 bytes. Any size other than these values will result in an error.

RX Config: There user must configure the RX parameters here on same places as in TX tab.

RX-Side Association Number (AN): The user must input the association number. By default, the value is 0. Possible values are between 0 to 3.

RX-Side Packet Number (PN): The user must input the packet number used by the cipher. By default, the value is 1. Possible values are all uint64_t values except 0 if XPN cipher is used, otherwise, all 32bit values are used except 0.

RX-Side Secure Association Key (SAK): The user must provide the SAK value. The possible values are between 0, 16 bytes, or 32 bytes. Any size other than these values will result in an error.

RX-Side Secure Channel Identifier (SCI): The user must provide the value of Secure Channel Identifier value. The possible values are between 0 and 8 bytes. Any size other than these values will result in an error.

RX-Side Short Secure Channel Identifier (SCCI): This field stores an unsigned integer of 32 bits. This can only be used in XPN ciphers. By default, the value is 0. There is no restriction on possible values.

RX-Side Salt: This field stores 12 bytes as a value for the Salt used in XPN ciphers. By default, the value is kept empty. Possible values are between 0 and 12 bytes. Any size other than these values will result in an error.

- Status Monitor

On the top of the Global Configuration panel, there is a drop-down section that shows the status of the device. The LEDs and DIP switches replicate the real behavior of the device as described in chapters: and 3.1.1 and 3.1.3

Additional information about the serial port status is also displayed:



Figure 5-22 Control Panel view

If desired, click *Hide Status* button to hide this section. To show it again, click button *Show Status*.

5.4.3 Control Panel

The Control Panel view is the following:

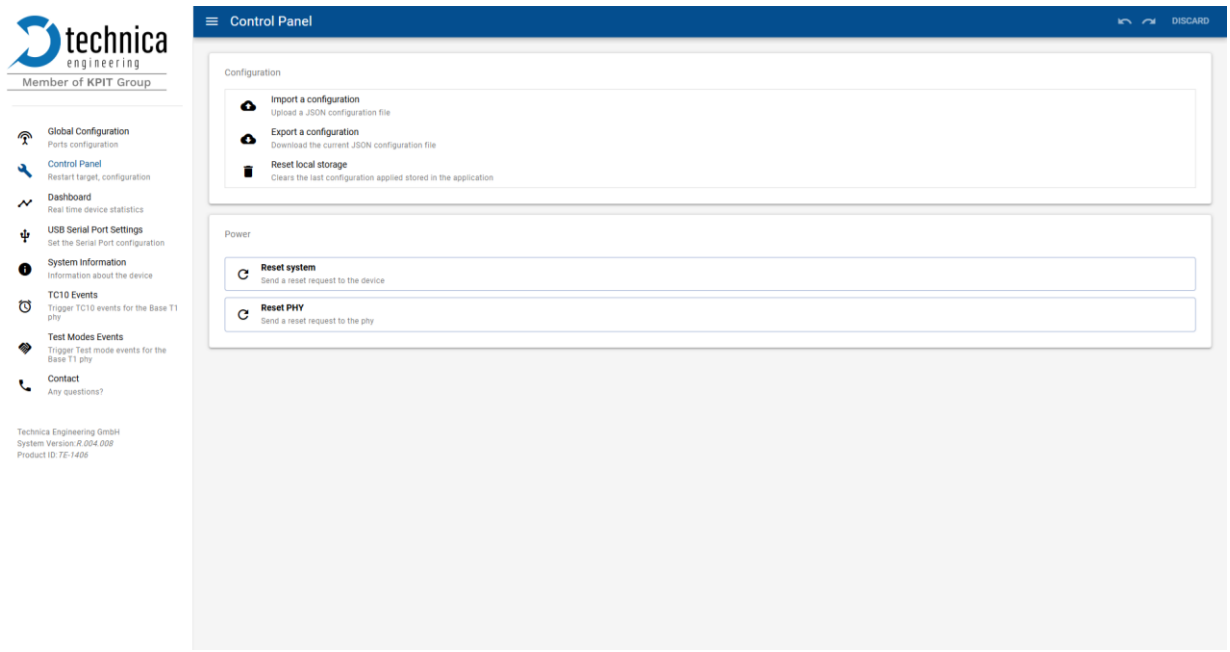


Figure 5-23 Control Panel view

The Control Panel allows Import and Export the device configuration and perform a reset system, which will reset all the MediaConverter or a reset PHY, which will assert a reset only to the BASE-T1 interface.

Asserting a system reset closes the Serial Port. Therefore, it needs to be reopened in the USB Serial Port Settings panel. Asserting a PHY reset causes a link down in BASE-T1 port.

5.4.4 Dashboard

This panel shows the counters packets of the BASE-T1 interface:

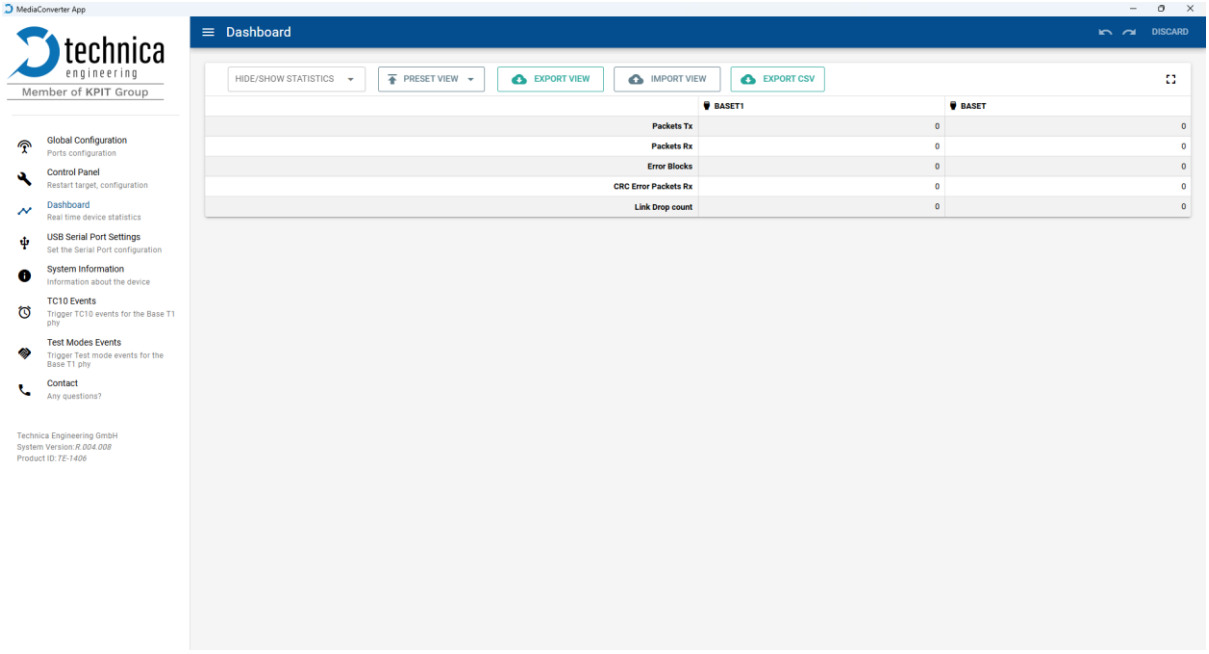


Figure 5-24 Dashboard panel view

There are several additional features in this panel:

- *Hide/Show Statistics* select: Allow users to hide a specific counter.
- *Preset View* select: Allow users to classify statistics in different groups. In this case, only *Rx/Tx Statistics* from BASE-T1 port are available.
- *Export View*: allow user to export the current view of the dashboard attending to the filters applied.
- *Import View*: allow user to import a dashboard view.
- *Export CSV*: allow user to export the statistics in a CSV file format.

5.4.5 System Information

The System Information panel is the following:

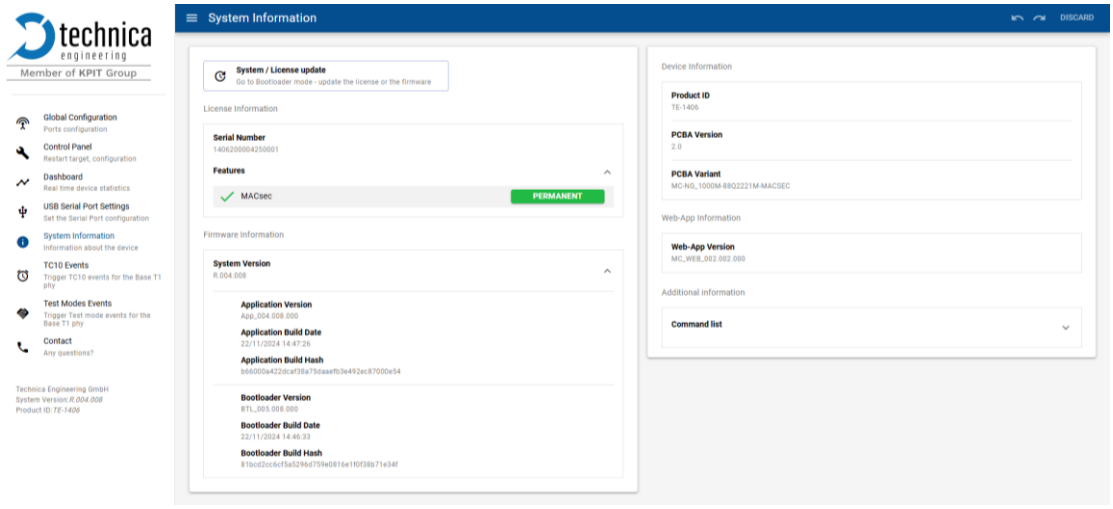


Figure 5-25 System Information view

This panel shows the system information of the MediaConverter, including product ID and PCBA information, the Serial Number and information about the release R.XXX.XXX.

There is a drop-down menu labelled *Command List* that shows all the available commands in the current release.

Refer to Firmware Update to see a detailed description of the Software Update process: 7.2

5.4.6 Test Modes Events

The web interface provides a panel to configure the Test Modes:

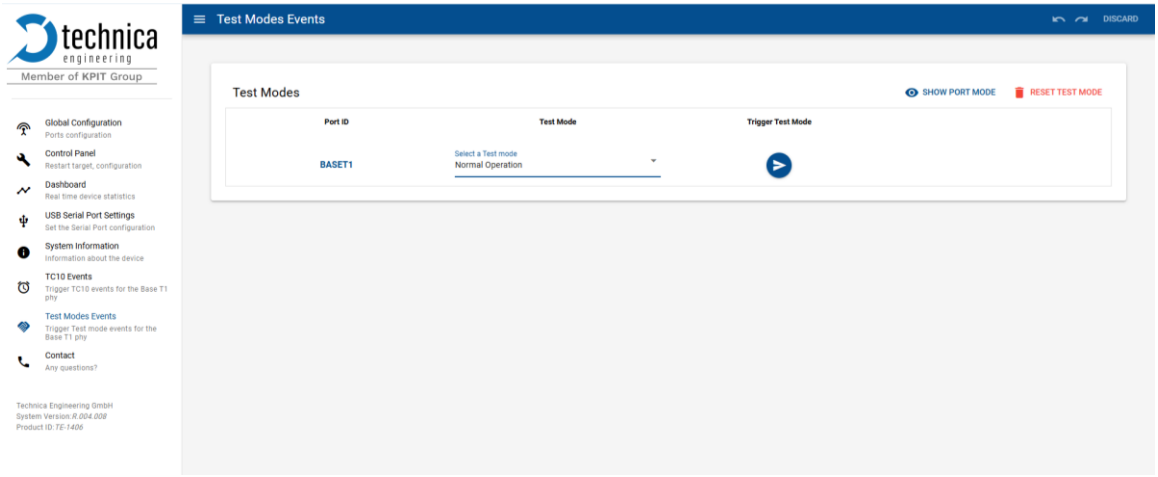


Figure 5-26 Test Modes Event panel

Attending to the IEEE 802.3 Specification, the test modes defined for 100MBPS are the following:

Test Mode	Description
Test Mode 0	Normal Operation
Test Mode 1	Maximum Transmitter Output Droop
Test Mode 2	Transmitter Timing Master Jitter and Clock Frequency
Test Mode 3	Reserved
Test Mode 4	Transmitter Distortion, MDI Return Loss and MDI Mode Conversion
Test Mode 5	Transmitter Power Spectral Density and Peak Differential Output
Test Mode 6	Reserved

Table 5-2 Tests Modes at 100MBPS

and for 1000MBPS:

Test Mode	Description
Test Mode 0	Normal Operation
Test Mode 1	Transmit Timing Jitter and clock frequency (Master/Slave)
Test Mode 2	Transmit MDI jitter test in MASTER mode
Test Mode 3	Reserved
Test Mode 4	Transmitter Distortion, MDI Return Loss and MDI Mode Conversion
Test Mode 5	Transmitter Power Spectral Density and Peak Differential Output
Test Mode 6	Maximum Transmitter Output Droop
Test Mode 7	Reserved

Table 5-3 Tests Modes at 1000MBPS

Select the desired configuration with the select component and click the blue button to set. To see which is the current test mode configured, click *"Show Port Mode"* button and to reset configuration, click *"Reset Test Mode"* button.

5.4.7 TC10 Events

The web interface provides a panel to trigger TC10 events: Sleep requests and Wakeup requests.

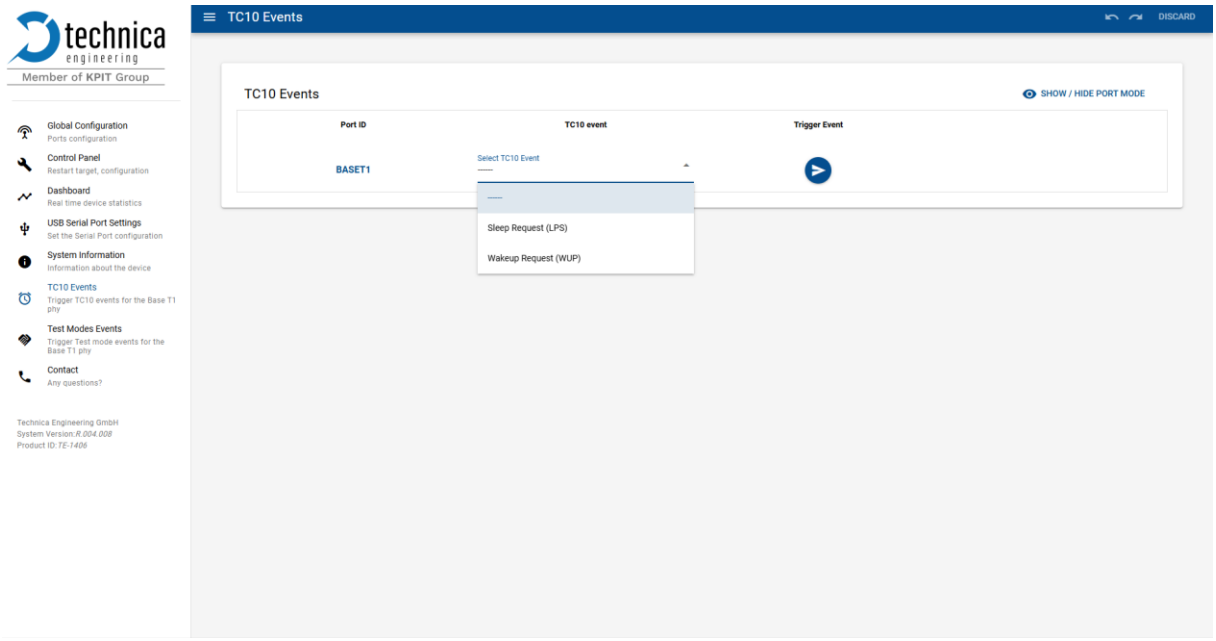


Table 5-4 TC10 Event panel

Select the desired event with the select component and click the blue button to trigger it. To see which is the current PHY mode status, either SLEEP mode or NORMAL mode, click *"Show Port Mode"* button.

5.4.8 Contact

In this page, contact information from Technica Engineering – GmbH can be found:

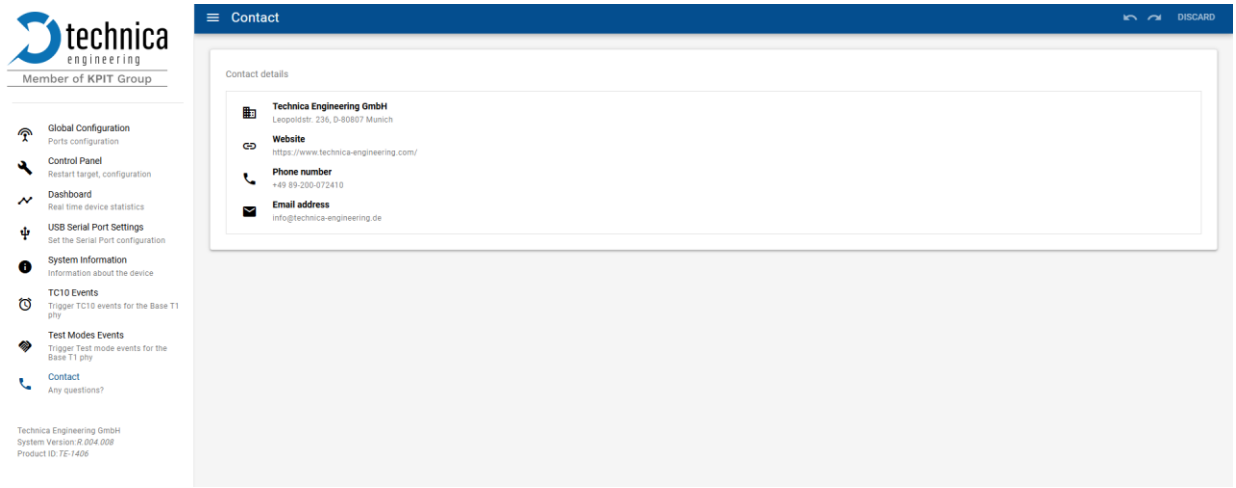


Table 5-5 Contact view panel

5.5 Device configuration

The WEB interface allows dynamically change the configuration of the device, in this case, the configuration parameters from the BASE-T1 interface. To do so, click to BASE-T1 port icon and refer to this section:

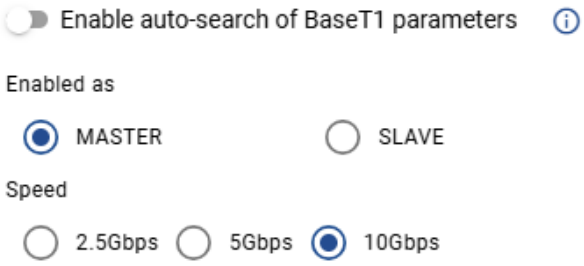


Table 5-6 BASE-T1 configuration parameters

Choose the parameters according to the desired configuration, then click *Save* button. Now, on the top-right of the screen, the following buttons will show:



Table 5-7 Apply Configuration button

The functionality of the buttons is described as follows:

- *Go back*: Removes the last configuration
- *Go forward*: Redo the last configuration
- *Discard*: Removes all the configured parameters
- *Apply Configuration*: Sets current configuration

When *Apply Configuration* is clicked, a dialog will appear:



Table 5-8 Apply Configuration dialog

Click *OK* to apply configuration to the device, *CANCEL* to cancel operation and *EXPORT CONFIG* to export the current configuration in a Json format file and apply configuration automatically.

5.6 License information

Some of the features available in the MediaConverters can be only available if there is a license active in the device. The list of all the features that are covered under the license is mentioned in this section. There can be two modes of license.

1. Unlicensed: No license is available for this feature
4. Licensed: License is available for this feature

MediaConverters TE-1405 and TE-1406 have the following features that need a license:

- MACsec feature

Command license:

License command allows to check the status of the license with all the available features.

Syntax:
- **license help:** Prints the license submenu.
- **license get:** Returns the status of all features licensed

Usage:
To get status of MACsec feature, use "**license get**". "MACsec feature: ENABLED" or "MACsec feature: DISABLED" according if the license is available or not.

6 Use cases

6.1 Standard use-case

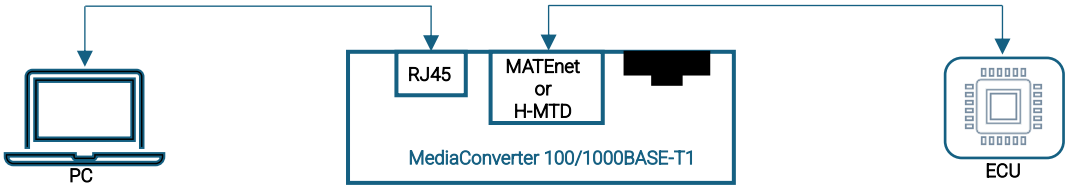


Figure 6-1 Standard use-case

- DIP Switch 1: The *Master/Slave* setting is dependent on the DUT. If DUT is Master, MC must be configured as Slave, DIP Switch 1 must be OFF [down], and vice versa
- DIP Switch 2: *100/1000 speed* setting is dependent on the speed of the DUT. If it is a 100BASE-T1 connection DIP Switch 2 must be OFF [down] and vice versa
- DIP Switch 3: *Auto Search feature* setting can be enabled if the configuration mode of the DUT is unknown. The MC will keep on changing its configuration until link up is detected.
- DIP Switch 4: *Frame Generator* function, always must be OFF [down]. Only in Frame Generator Mode, the DIP Switch must be ON [up], in this case, the device no longer works as a converter

7 Firmware update

To perform the firmware update of the device is necessary to download current FW file from Customer portal of Technica Engineering, link: https://portal.technica-engineering.de/downloads

Firmware update of MediaConverter is possible to do in two ways. The first possibility is to update MediaConverter via serial interface (console) and second via MediaConverter GUI PC application. In this chapter both ways for update procedures are described.

7.1 FW update via console

To connect PC with MediaConverter, follow the steps described in chapter: 4.3.

The bootloader needs to be flashed before flashing the firmware file.

1. Bootloader release R.004.XXX

The version is mentioned in the title screen when the device is accessed serially. If the user wants to explicitly check the bootloader version, the command “version” can be used. Below is the output of the “version” command.

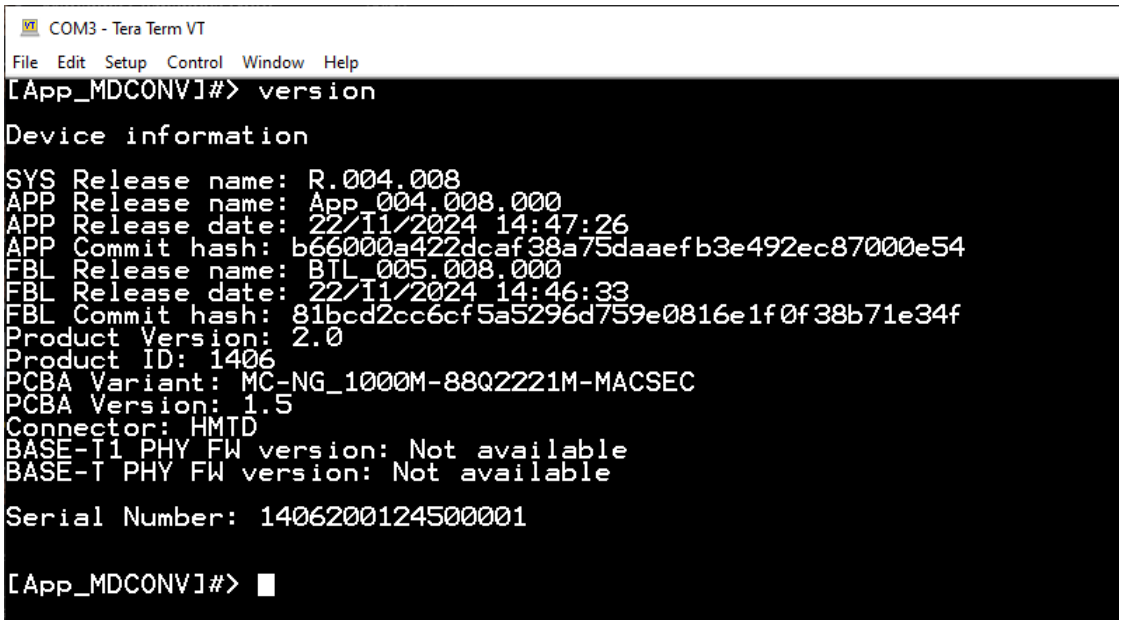


Figure 7-1 Output of "info" command

2. At first is needed to unpack .7z file and place the bootloader file and firmware file to place (e.g. desktop), where can user find it quickly
3. If the bootloader version is BTL_004.XXX, run the bootloader updater file named: bootloader_updater_signed.tefw
4. Enter “fbl” to go to bootloader mode. It will appear as below:



Figure 7-2 Bootloader view

5. Type “flash” so that you will have a few seconds to perform the next step.



Figure 7-3 Information from console

6. Select “File” →” Transfer”→” XMODEM”→” Send” and then select the provided firmware file. It should have the ‘.tefw’ extension.

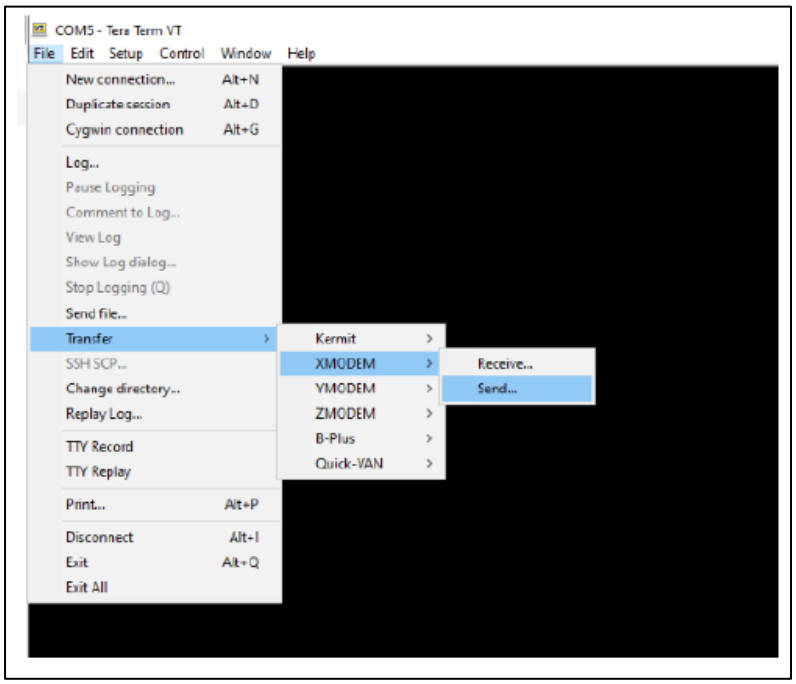


Figure 7-4 Bootloader flashing procedure

7. A window with the progress bar will appear as below:

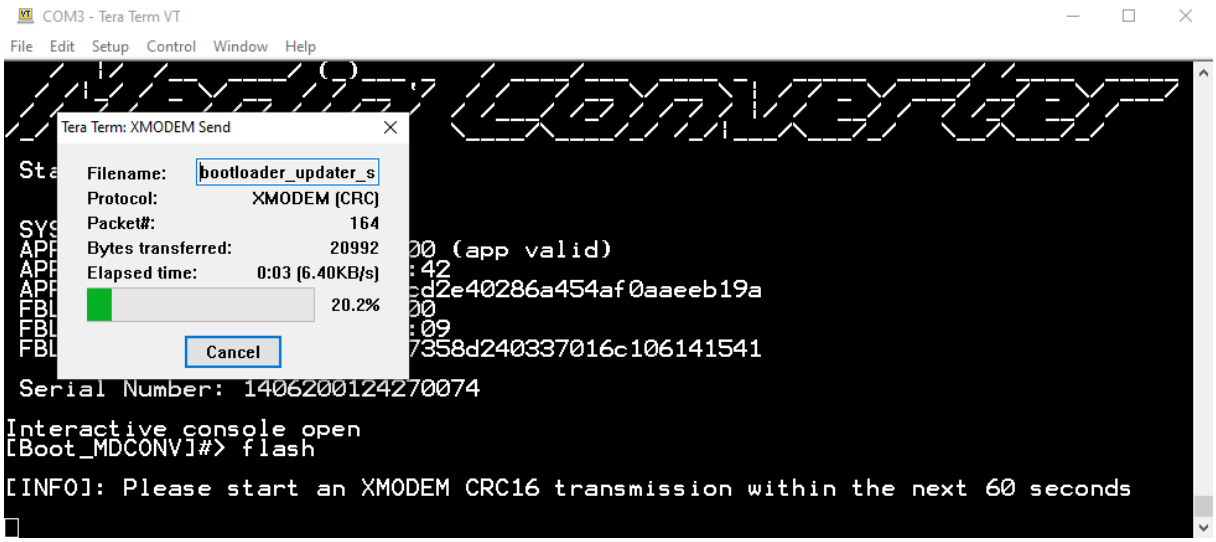


Figure 7-5 Flashing bar

7.1.1 Firmware flashing

After successful bootloader flashing, is necessary to do same process for firmware update.

1. Enter “fbl” to go to bootloader mode. It will appear as below: (if the bootloader step has not been done, if it is done, ignore this step).



Figure 7-6 Bootloader view

2. Type “flash” so that you will have 60 seconds to perform the next step.
3. Select “File” →” Transfer”→” XMODEM”→” Send” and then select the provided firmware file. Name of FW is: firmware_full_signed.tefw
4. A window with the progress bar will appear.
5. The firmware update will be completed once the progress bar reaches 100%. A message indicating that the procedure has been performed successfully will appear in the console window.

```
[INFO]: Flashing successful
```

Figure 7-7 Information: flashing successful

6. If the device does not exit out of bootloader mode, type the command “app” to manually exit from bootloader mode. You can also confirm if the device is out of bootloader mode through the “GP” LED. It should be continuously lit.

7.2 Firmware flashing via MediaConverter PC App

To update the firmware via MediaConverter App, follow the steps in App GUI:

1. Go to System Information panel and click System/License Update.
2. A dialog box will appear, click Ok.
3. The following dialog will appear:

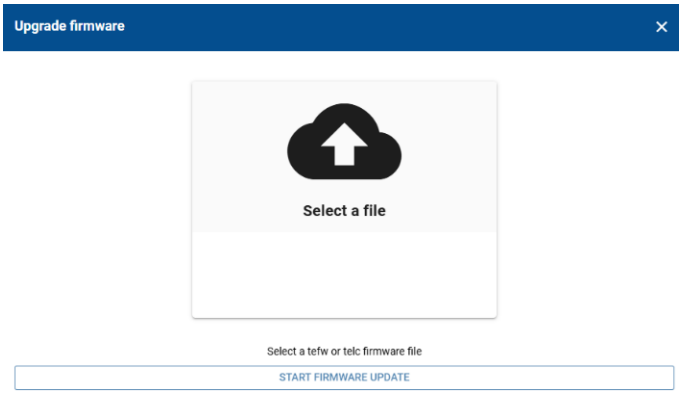


Figure 7-8 Upgrade firmware view without file

4. To upload a file, click in Select a file and navigate through your files. Only .tefw files will be accepted.

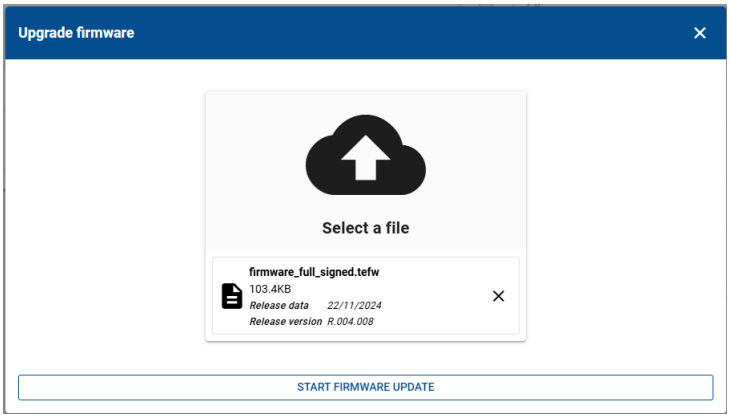


Figure 7-9 Upgrade firmware view with file

Drag and drop feature is for now not available on this interface

5. Then, click Start Firmware Update button.
6. A loading page with the message “Software Update Ongoing. Expected time: XX seconds” will appear. Wait until It finishes.
7. If successful, no error message appears, and a pop-up dialog will appear asking if another file want to be uploaded or reset device want to be asserted and return to application.
8. If some error comes up during the software update, the message will be shown:

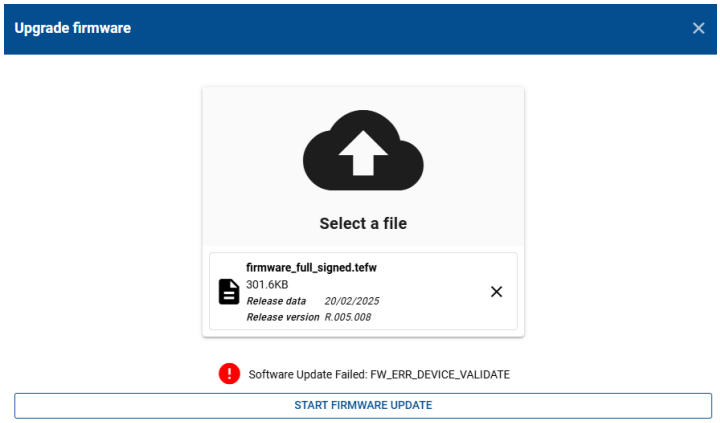


Figure 7-10 Software Update error

This means that the firmware file is not the correct one, please try again with correct file.

7.2.1 Update Errors

During the software update, some errors may occur. This section provides a list of possible errors along with their corresponding solutions.

- *FW_ERR_BOOTLOADER_TOO_OLD:* Bootloader version is too old. Therefore, it needs to be updated. To do so, upload the *bootloader_updater.tefw* file following the same steps described in section 1.6.
- *FW_ERR_PRODUCT_MISMATCH:* The *.tefw* file does not correspond to the targeted device. You must assure that the file applies to your device.
- *FW_ERR_FIRMWARE_TOO_OLD:* The *.tefw* file corresponds to older System Releases, thus it will not be possible to go back. Be sure to always use the last release available.
- *FW_ERR_UNRECOGNISED_FILE_FORMAT:* The *.tefw* file is not recognized by the MediaConverter. Be sure to use always the files provided by Technica without any changes.

7.2.2 Firmware Updates from not supported device

In case a not supported device is available, the web interface allows to update the firmware.

When opening a not supported device, the following message will appear to indicate so:

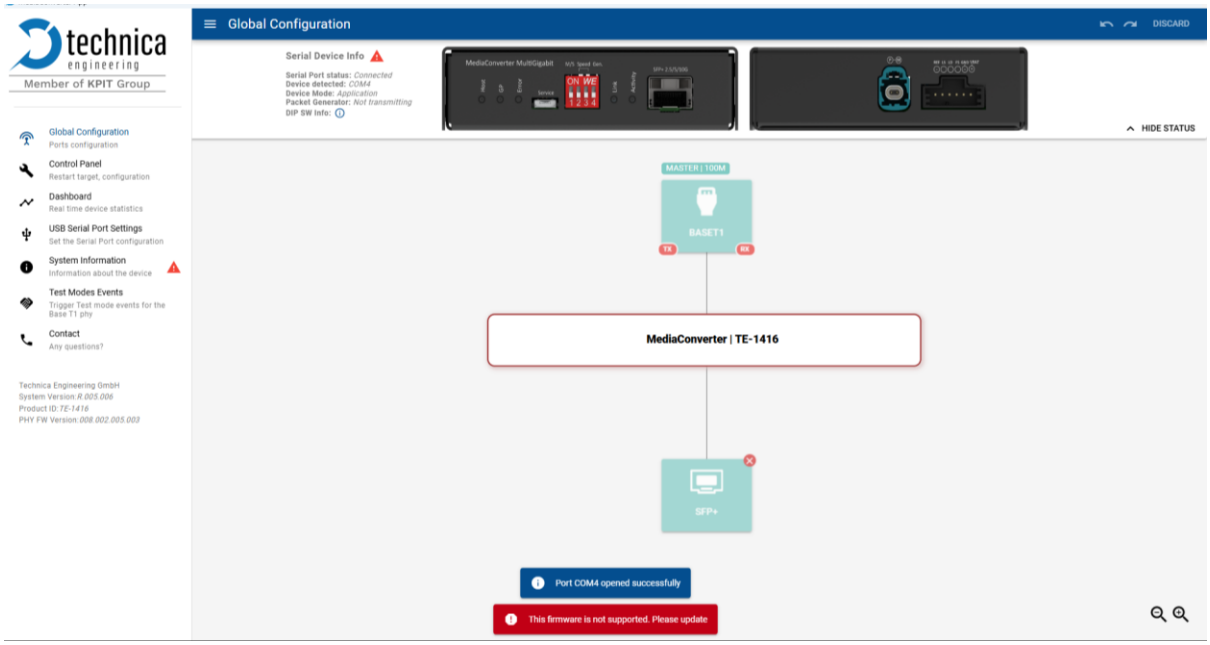


Figure 7-11 Global configuration for not supported device

The web interface will only allow the user to go to *System Update* in *System Information Panel*. The buttons of both ports will be disabled, invalidating the chance to apply any further configuration before updating the software to a supported device.

8 Additional information

- The 100/1000BASE-T1 MediaConverter is optimized for automotive use. The maximum cable length for 100/1000BASE-T1 segments is limited to 15 meters.
- It is not possible to mix up different speeds from RJ-45 to BASE-T1-Port
- Do not use it as a *Frame Generator* while data conversion. Data loss may occur.

9 Declaration of conformity/ Konformitätserklärung

English:

Hereby, Technica Engineering GmbH declares that the product 100/1000BASE-T1 MediaConverter TC10 (TE-1405 or TE-1406), complies with Directive 2015/863/EU and 2014/30/EU. The full text of the EU declaration of conformity is available at the following internet address:

https://www.technica-engineering.de/

or on Customer portal:

https://portal.technica-engineering.de/downloads

Deutsch:

Hiermit erklärt Technica Engineering GmbH, dass das Produkt 100/1000BASE-T1 MediaConverter TC10 (TE-1405 oder TE-1406) die Richtlinie 2015/863/EU and 2014/30/EU entspricht. Der vollständige Text der EU-Konformitätserklärung ist unter der folgenden Internetadresse verfügbar:

https://www.technica-engineering.de/

oder im Kundenportal:

https://portal.technica-engineering.de/downloads

10 Important notes

10.1 General operating and safety strategy of Technica Engineering's Products

Technica Engineering's products are designed for operation in automotive systems and for supply voltage of typical 12 V (voltage range 6V - 30V). The applicable limit values adhere to the standard norms automotive onboard power systems correspondingly and can be found in the mentioned norms.

Should Technica Engineering's products be operated in voltage ranges beyond those specified in the norms, which represents a breach of the conditions of operation, then this will void the product warranty and Technica Engineering will assume no liability whatsoever for the results and/or consequences thereof.

This is especially valid whenever the voltage level reaches or exceeds the limits of the low-voltage directive. In this case, damage to the devices cannot be excluded. Due to the manufacturing characteristics of the devices, there is no imminent fire hazard from the device itself, if the devices are being operated in an environment according to the conditions of use. A secondary fire hazard cannot be excluded, should those conditions not be met. Protection against overvoltage cannot be provided in such a breach of the conditions of use.

10.2 General design rules for the power supply of Technica Engineering's products

The power supply solution of Technica Engineering's products is equipped with a self-protection mechanism. This automatic function protects the internal electronic components against excessive temperature and too high a supply voltage by switching the device off. This automatic switch-off function is independent of any software function. Excessive temperatures in the device can be caused by high environmental temperatures or internal failures of the device. The device can protect itself by shutting down its power supply avoiding further damage to itself.

The operator/user of these devices must take measures for the right protection of the power supply lines and the connector. The maximum power consumption of the Technica Engineering products is written in this user manual.

Please consider that some of the products work with a step-up converter. So, the current needed in case of undervoltage conditions must be considered higher than a typical voltage level. The user must choose the right fuse and/or other over current protection measures.

10.3 Warranty and Safety Information



Before operating the device, please read this User Manual thoroughly and follow the provided information and instructions. Keep the manual for future reference.



Operate the device only as instructed in this manual.
Ensure the device is used exclusively in dry conditions.
Refrain from inserting any foreign objects into the device's slots or openings.
Do not use or power a device that is damaged or malfunctioning.



Please do not attempt to open or disassemble the device. Unauthorized tampering or modifications will result in the immediate voiding of the warranty and may also compromise the safety and functionality of the device.



This product is intended for use in automobiles or similar environments, including office, laboratory, and workshop test setups. Technica Engineering products are not designed for standard IT use but as specialized test modules for experts in development and testing. Ensure adequate ventilation and proper external power supply with appropriate cable protection (e.g., fuses) when integrating these modules. Maximum current load specifications are detailed in the user manual. Technica Engineering products should be included in safety assessments and are aligned with standard Quality Management (QM) per ISO 26262.



Do not cover the device to prevent fire hazards.
Keep the device away from highly flammable materials to avoid fire risk.
Do not operate the device above the specified temperature limits.
Note that the operating temperature refers to the ambient temperature of the installation area.



This symbol is applicable only within the European Union. To discard this product, please consult your local authorities or dealer for proper disposal methods. Technica Engineering GmbH is registered as a manufacturer of the "Technica Engineering" brand, specializing in "Small devices of Information and Telecommunications Technology for exclusive use in non-private households." WEEE Registration Number: DE 20776859.



This product has been manufactured and tested to comply with CE (Conformité Européenne) standards, as indicated by the EU Declaration of Conformity.

11 Wichtige Hinweise

11.1 Allgemeine Betriebs- und Sicherheitsstrategie für Produkte von Technica Engineering

Die Produkte von Technica Engineering sind für den Einsatz in Fahrzeugsystemen und für eine typische Versorgungsspannung von 12 V (Spannungsbereich 6V - 30V) ausgelegt. Die geltenden Grenzwerte entsprechen den Normen der Bordnetze von Kraftfahrzeugen und sind in den genannten Normen zu finden.

Sollten die Produkte von Technica Engineering in Spannungsbereichen betrieben werden, die außerhalb der in den Normen angegebenen Bereichen liegen, was einen Verstoß gegen die Betriebsbedingungen darstellt, erlischt die Produktgarantie und Technica Engineering übernimmt keinerlei Haftung für die daraus resultierenden Ergebnisse und/oder Folgen.

Dies gilt insbesondere dann, wenn das Spannungsniveau die Grenzwerte der Niederspannungsrichtlinie erreicht oder überschreitet. In diesem Fall kann ein Schaden an den Geräten nicht ausgeschlossen werden. Aufgrund der Herstellungseigenschaften der Geräte besteht keine unmittelbare Brandgefahr durch das Gerät selbst, sofern die Geräte in einer Umgebung gemäß den Nutzungsbedingungen betrieben werden. Eine sekundäre Brandgefahr kann jedoch nicht ausgeschlossen werden, falls diese Bedingungen nicht eingehalten werden. Ein Schutz gegen Überspannung kann in einem solchen Fall von Verstoß gegen die Nutzungsbedingungen nicht gewährleistet werden.

11.2 Allgemeine Designregeln für die Stromversorgung von Produkten von Technica Engineering

Die Stromversorgungslösung der Produkte von Technica Engineering ist mit einem Selbstschutzmechanismus ausgestattet. Diese automatische Funktion schützt die internen elektronischen Komponenten vor übermäßiger Temperatur und zu hoher Versorgungsspannung, indem das Gerät abgeschaltet wird. Diese automatische Abschaltfunktion ist unabhängig von jeglichen Softwarefunktionen. Übermäßige Temperaturen im Gerät können durch hohe Umgebungstemperaturen oder interne Fehler des Geräts verursacht werden. Das Gerät kann sich durch Abschalten seiner Stromversorgung selbst schützen und so weiteren Schaden vermeiden.

Der Betreiber/Nutzer dieser Geräte muss Maßnahmen ergreifen, um den richtigen Schutz der Stromversorgungsleitungen und des Steckverbinders sicherzustellen. Der maximale Stromverbrauch der Produkte von Technica Engineering ist in diesem Benutzerhandbuch angegeben.

Bitte beachten Sie, dass einige der Produkte mit einem Step-Up-Konverter arbeiten. Daher muss berücksichtigt werden, dass der Strombedarf im Falle von Unterspannung höher sein kann als bei einem typischen Spannungsniveau. Der Nutzer muss die richtige Sicherung und/oder andere Überstromschutzmaßnahmen auswählen.

11.3 Garantie- und Sicherheitsinformationen



Bevor Sie das Gerät in Betrieb nehmen, lesen Sie bitte dieses Benutzerhandbuch sorgfältig durch und befolgen Sie die angegebenen Informationen und Anweisungen. Bewahren Sie das Handbuch als zukünftige Referenz auf.



Betreiben Sie das Gerät nur gemäß den Anweisungen in diesem Handbuch.
Stellen Sie sicher, dass das Gerät ausschließlich unter trockenen Bedingungen verwendet wird.
Vermeiden Sie es, Fremdkörper in die Schlitze oder Öffnungen des Geräts einzuführen.
Verwenden oder betreiben Sie kein Gerät, das beschädigt oder defekt ist.



Bitte versuchen Sie nicht, das Gerät zu öffnen oder auseinanderzubauen. Unbefugte Manipulationen oder Modifikationen führen zum sofortigen Erlöschen der Garantie und können zudem die Sicherheit und Funktionalität des Geräts beeinträchtigen.



Dieses Produkt ist für den Einsatz in Automobilen oder ähnlichen Umgebungen bestimmt, einschließlich Büro-, Labor- und Werkstatt-Testaufbauten. Produkte von Technica Engineering sind nicht für den Standard-IT-Einsatz konzipiert, sondern als spezialisierte Testmodule für Experten in der Entwicklung und Prüfung. Sorgen Sie für eine ausreichende Belüftung und eine ordnungsgemäße externe Stromversorgung mit geeigneter Kabelsicherung (z.B. Sicherungen) beim Einbau dieser Module. Die maximalen Stromlastspezifikationen sind im Benutzerhandbuch detailliert beschrieben. Produkte von Technica Engineering sollten in Sicherheitsbewertungen einbezogen werden und sind mit dem Standard-Qualitätsmanagement (QM) gemäß ISO 26262 abgestimmt.



Bedecken Sie das Gerät nicht, um Brandgefahren zu vermeiden.
Halten Sie das Gerät von leicht entflammbaren Materialien fern, um Brandrisiken zu vermeiden.
Betreiben Sie das Gerät nicht außerhalb der angegebenen Temperaturgrenzen.
Beachten Sie, dass sich die Betriebstemperatur auf die Umgebungstemperatur des Installationsbereichs bezieht.



Dieses Symbol ist nur innerhalb der Europäischen Union gültig. Um dieses Produkt zu entsorgen, konsultieren Sie bitte Ihre örtlichen Behörden oder Ihren Händler für eine ordnungsgemäße Entsorgungsmethode. Technica Engineering GmbH ist als Hersteller der Marke „Technica Engineering“ registriert und spezialisiert auf „Kleingeräte der Informations- und Telekommunikationstechnologie zur ausschließlichen Nutzung in nicht-privaten Haushalten“. WEEE-Registrierungsnummer: DE 20776859.



Dieses Produkt wurde hergestellt und getestet, um den CE-Standards (Conformité Européenne) zu entsprechen, wie durch die EU-Konformitätserklärung angegeben.

11.4 IMPRINT

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